

Review

A comprehensive review of object detection with traditional and deep learning methods

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ABSTRACT

Object detection is one of the most important and challenging tasks of computer vision. It has numerous applications in the fields of agriculture, defence, retail markets and manufacturing units, transportation, social media platforms, medical, wildlife monitoring and conservation. This survey aims to give researchers a comprehensive understanding of the current state of object detection algorithms. In this review, object detection and its different aspects have been covered in detail. This review paper starts with a quick overview of object detection followed by traditional and deep learning models for object detection. The section on deep learning models provides a comprehensive overview of one-stage and two-stage object detectors. A detailed discussion is given of the transformer-based detectors and lightweight networks category. Additionally, the evaluation metrics used for object detection methods are discussed systematically. The best object detection algorithms for different applications are discussed at the end of the survey. This survey is useful for beginners who want to study different object detection algorithms and their use in different applications.

1. Introduction

Object detection is one of the most important tasks in computer vision, involving the identification of visual instances of specific object classes within an image or complex real-world scenes. Its ability to identify, classify, and localize multiple objects within an image or video frame makes it a vital component of numerous applications. Object detection is crucial across industries, enabling autonomous navigation, security surveillance, medical imaging, retail automation, precision farming, and quality control in manufacturing. Recognizing objects in real-time, it enhances safety, efficiency, and decision-making. Advancements in AI and deep learning continue to drive its transformative impact, making it indispensable for the future. Due to its wide range of applications and recent technological advancements, this field has gained significant attention. The primary objective of object detection methods is to develop computational models that provide essential data for computer vision-based systems. Currently, object detection is a deeply researched area, both in academic studies and practical applications [17].

The Viola-Jones detector, Histogram of Oriented Gradients (HOG), and other manually constructed feature extractors formed the early

object detection models, also referred to as traditional models. These models performed poorly on unknown datasets and were slow and unreliable. The field of visual perception transformed with the reintroduction of convolutional neural networks (CNNs) and deep learning for image classification. Currently, deep learning is being used in a variety of computer vision applications, including semantic segmentation, detecting objects, image classification, and image retrieval. Lightweight neural networks or mobile-friendly neural networks are deep neural networks which are designed for mobile devices or other resource-limited hardware [135].

Several reviews on object detectors have been published in recent years [131,133,140,146–151], serving as early efforts to summarize the object detection models available at the time of their publication. However, the rapid advancements in computer vision, particularly in object detection, have led to the emergence of numerous enhanced deep learning models. This study presents a comprehensive analysis of object detection methodologies, encompassing traditional approaches, deep learning-based techniques, and their subcategories, including one-stage, two-stage, transformer-based, and lightweight models. It examines their advantages, limitations, and the evolution of network architectures, with a particular emphasis on the research gap in lightweight models—a

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topic that has not been explored in such depth in previous reviews. This survey provides a comprehensive review of traditional object detection methods as well as deep learning-based detectors, which are further classified into two-stage methods, one-stage methods, transformer-based detectors, and lightweight networks, each offering distinct advantages for various object detection applications. The main contributions of this paper are as follows:

1. This work provides an in-depth analysis/discussion of different object detection methods, like traditional, deep learning methods where they are further categorised into one and two-stage methods, transformer-based and lightweight networks.
2. Traditional methods are analysed and based on analysis the advantages and disadvantages are discussed in the paper.
3. The analysis of deep learning methods is discussed and highlights the changes in network designs and their implication on the performance of detectors.
4. This work discusses in detail the two-stage and one-stage models, transformer-based models and lightweight models. Although lightweight model research has progressed, there aren't many reviews of it. The goal of this endeavour is to bridge the gap.
5. Comparative analysis of object detection algorithms based on strengths and weaknesses, the best object detection methods for different applications are discussed and highlight the metrics used for the evaluation of the methods. That will help beginner object detection researchers.
6. The paper ended with a conclusion along with future directions and key challenges in object detection.

Here we systematically reviewed various object detection architectures and their associated technologies, as illustrated in Fig. 1. The paper is organized as follows. In Section 2, a summary of previous survey papers on object detection is discussed. Section 3 explains the need and motivation for having this survey. Section 4 is divided into four major sections, each studying a different category of object detectors. The different major metrics used for the evaluation of object detection methods are discussed in Section 5. Section 6 presents a comparative analysis of various object detection algorithms, evaluating their strengths and weaknesses to identify the most effective models and based on our survey, we highlight the best-performing algorithms suited for different applications. In Section 7, we provide a comprehensive conclusion, summarizing key insights and recommending the optimal object detection approaches based on accuracy, speed, and real-world applicability.

2. Overview of object detection algorithms

Object detection has emerged as a cornerstone in the field of computer vision, enabling machines to identify and localize objects within images or videos. Over the years, numerous algorithms have been developed, each offering unique approaches to improve accuracy, speed, and scalability. Classical methods such as Viola-Jones and HOG-SVM laid the foundation by employing hand-crafted features and sliding window techniques. The advent of deep learning revolutionized object detection with the introduction of convolutional neural networks (CNNs), giving rise to modern architectures like R-CNN, SSD, and YOLO, which prioritize real-time performance and precision. The several methods and algorithms employed for different object detection applications are discussed in Table 1.

3. The need and motivation for having this survey

Object detection is a fundamental computer vision task that involves identifying and locating objects within an image or video. The need for object detection models arises from their diverse real-world applications, as well as the continuous advancements in technology that make

these tasks possible. The motivation for doing this survey is mentioned below.

- **Automation:** The demand for automation in diverse industries drives the need for object detection models, enabling machines to perceive and interact with their surroundings intelligently. In autonomous vehicles like drones, cars etc. object detection models are required to identify pedestrians, vehicles, road signs, and obstacles in real time to navigate safely [89].
- **Healthcare:** In medical imaging, object detection is used to locate tumours, lesions, and abnormalities in X-rays, CT scans and MRIs, aiding in diagnostic processes.
- **Surveillance:** Models identify faces in surveillance footage, helping law enforcement and security personnel monitor public spaces, identify suspects, or secure access points and monitor real-time video streams to detect suspicious activities, identify intruders, or track moving objects in public spaces or homes [18].
- **Sports:** Object detection can be used to track players, analyse their movements, and provide insights into game strategies by analysing live broadcasts.
- **Threat Detection:** Detecting weapons, explosives, or other dangerous objects in X-ray images or CCTV footage at airports, train stations, and other public places is critical for preventing security threats [19].
- **Wildlife Conservation:** Object detection is used in camera traps or drones to monitor wildlife populations and track endangered species, contributing to conservation efforts.
- **Disaster Management:** In post-disaster scenarios (e.g., earthquakes, floods), object detection can help locate survivors, detect damaged structures, and monitor affected areas [135].
- **Agriculture:** Precision agriculture utilizes object detection to monitor crop health, detect pests, and manage resources more efficiently, contributing to sustainable farming practices [135].

Despite the advances, there are significant challenges that motivate the continuous survey and development of object detection models. The first key challenge is achieving high-quality detection across multiple scales. This means detecting both large and small objects simultaneously with the same model, which remains a difficult task. Secondly, handling complex scenes which includes detecting objects in crowded or cluttered environments, such as traffic scenes, supermarkets, or gatherings, is complex and the third challenge is to reduce false positives/negatives, while object detection models have become more accurate, reducing the rates of false positives (incorrect detections) and false negatives (missed detections) is still a priority, especially in high-stakes applications like healthcare or autonomous driving. The need for object detection models is driven by a wide array of practical applications, ranging from automation and real-time analysis to safety, security, and societal benefits. The motivation for developing increasingly sophisticated and efficient models lies in addressing the challenges of real-world environments, enabling AI systems to perceive and interact with the world more effectively [135]. Continuous advancements in machine learning and deep learning architectures are essential to meet these growing demands.

4. Object detection methods

This section is divided into two types of detectors: traditional methods and deep learning methods. The deep learning methods are further categorized into single-stage, two-stage, transformer-based approaches, and lightweight networks, as shown in Fig. 1. A two-stage detector is a network that has a separate module that generates region proposals. In the first stage, these models attempt to identify an arbitrary number of item proposals in an image; in the second stage, they classify and place those things [135]. These systems typically take longer to generate proposals, have more complex architectures, and lack global

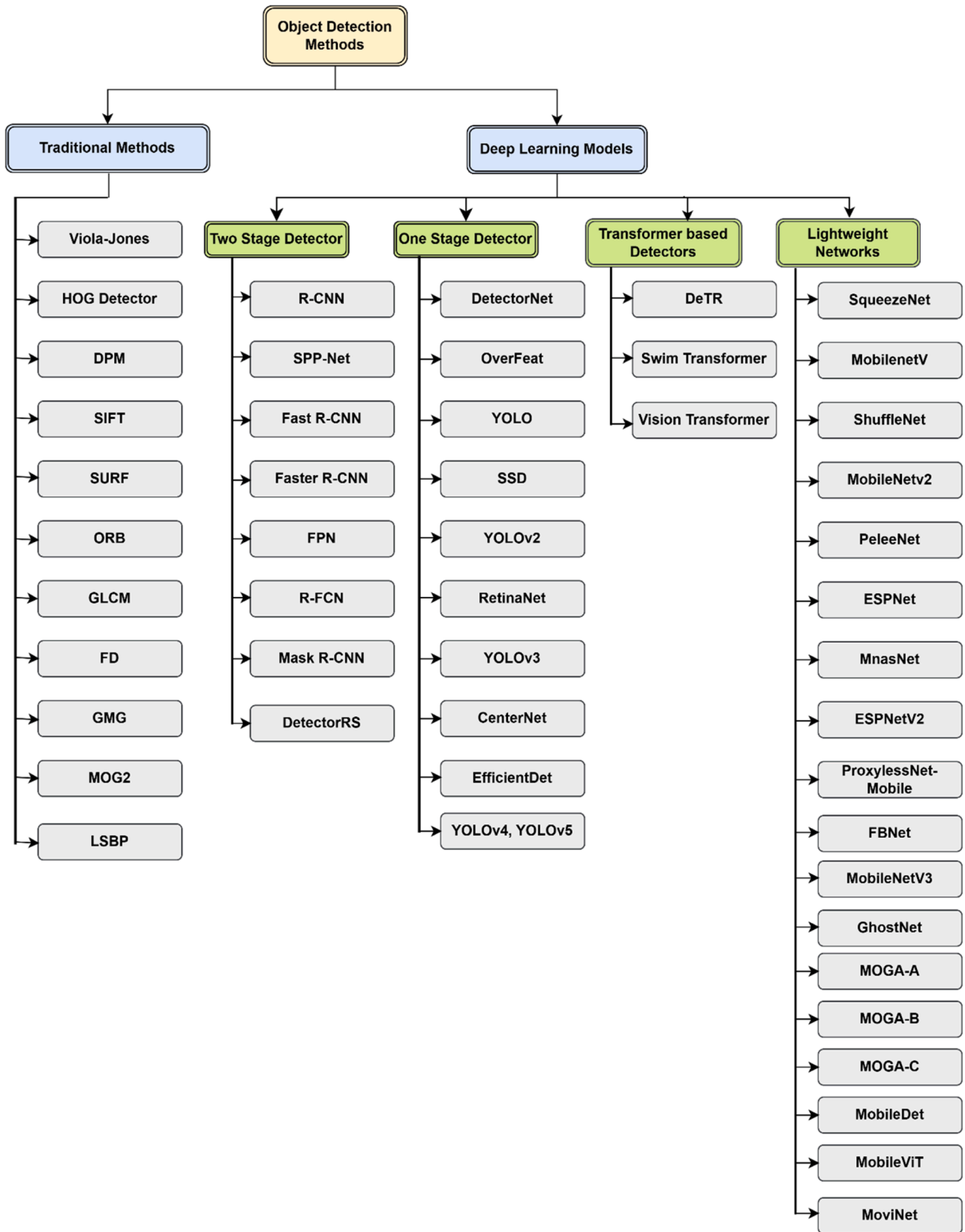


Fig. 1. Overview of object detection methods.

Table 1

Overview of object detection algorithms for different applications.

Sr. No.	Year	Algorithm/ Method	Discussion	Refs.
	2017	Histogram of Oriented Gradients (HOG) and linear Support Vector Machine (SVM)	The authors propose an algorithm designed for vehicle detection in urban environments, which plays a crucial role in driver assistance systems and autonomous driving. The results demonstrate the algorithm's robustness and accuracy across various scenarios.	[4]
	2017	Oriented FAST and Rotated BRIEF (ORB)	The authors investigate the impact of prefiltering on Oriented Fast and Rotated BRIEF (ORB)-based object detection. They also propose a covariance-based approach to determine Hamming distance thresholds for each prefiltered ORB detector, comparing the minimum Hamming distance values between positive and negative training images.	[13]
	2018	Fast/Faster R-CNN	This paper illustrates the application of fully convolutional architectures within Fast/Faster R-CNN frameworks. Experimental results reveal that using a convolutional layer-based classifier significantly enhances object detection performance compared to a fully connected layer-based classifier. Furthermore, employing deeper ConvNets as feature classifiers leads to improved detection accuracy.	[25]
	2018	Histogram of Oriented Gradients (HOG) and linear Support Vector Machine (SVM)	This research focuses on human object detection and tracking, addressing challenges in complex conditions. The proposed model introduces a novel approach to human object detection using a cluster segmentation method. Features are extracted based on the HOG and classification is performed using the Support Vector Machine (SVM) algorithm.	[3]
	2019	Viola-Jones algorithm	This paper presents an enhanced Viola-Jones algorithm for face detection, optimized for HoloLens. The improvement builds upon the classical Viola-Jones approach by expanding Haar-like rectangular features to boost detection efficiency. Additionally, it incorporates two-dimensional convolution separation and an image re-sampling technique to accelerate the detection process.	[1]
	2019	An Augmented Linear Mixing Model	This paper introduces the Augmented Linear Mixing Model (ALMM) to enhance hyperspectral unmixing by addressing spectral variability, a key limitation of the traditional Linear Mixing Model (LMM). ALMM incorporates scaling factors	[156]

Table 1 (continued)

Sr. No.	Year	Algorithm/ Method	Discussion	Refs.
			and other spectral variations using a low-coherence dictionary learning approach, ensuring minimal correlation with end member signatures. Optimized with the Alternating Direction Method of Multipliers (ADMM), ALMM outperforms existing methods like PLMM and ELMM, delivering more accurate abundance estimations and improved spectral variability handling in hyperspectral imagery.	
	2020	Speeded-Up Robust Features (SURF)	3D reconstruction using a single-antenna imaging radar faces challenges in correlating scattering centres. To address this, a method is proposed for associating scattering centres by obtaining the transformation matrix between adjacent image frames through rough matching of Speeded-Up Robust Features (SURF). The association results of the scattering centres are then determined by minimizing the 2-norm after back-projection.	[12]
	2020	Deformable Part Models	Part-level representations are crucial for robust person re-identification. However, inaccuracies in pedestrian detection often lead to significant misalignment of body parts, compromising the quality of part representations. To address this issue, the authors propose a novel model called the Convolutional Deformable Part Model (CDPM). The results highlight the effectiveness of CDPM in achieving accurate part alignment.	[5]
	2020	Gray Level Co-occurrence Matrix (GLCM)	This study concentrates on the preprocessing, feature extraction, and selection phases for ultrasound images of the kidney to develop a classification model. Four preprocessing techniques—cropping, interpolation, rotation, and background removal are applied to improve image quality and facilitate more accurate and effective diagnosis. Next, a range of second-order statistical texture features, including energy, entropy, homogeneity, correlation, contrast, and dissimilarity, are extracted using the Gray Level Co-occurrence Matrix (GLCM).	[15]
	2020	Spatial Pyramid Pooling (SPP-Net)	This paper proposes a method using the SPP-Net model, which can accept images of various sizes as input, including colour images, offering a wide range of features for detecting variants.	[23]

(continued on next page)

Table 1 (continued)

Sr. No.	Year	Algorithm/ Method	Discussion	Refs.
	2020	Region-based Fully Convolutional Network (R-FCN)	The authors propose an improved multidimensional region-based fully convolutional network (mRFCN) for an automated decision support system aimed at lung nodule detection and classification.	[29]
	2020	Mask Region-based Convolutional Neural Network (Mask R-CNN)	Traffic congestion caused by vehicular accidents significantly disrupt normal travel, necessitating the development of accurate and effective mitigation methods. To address traffic accident compensation issues promptly, this paper proposes a vehicle-damage-detection segmentation algorithm based on transfer learning and an enhanced Mask Regional Convolutional Neural Network (Mask R-CNN).	[30]
	2020	Fourier-based Rotation-Invariant Feature Boosting (FRIFB)	FRIFB is a novel framework for geospatial object detection in remote sensing, addressing deformations from scaling and rotation. It leverages FourierHOG for rotation-invariant features, refines them with ACF, and optimizes testing with FPGM to reduce computational costs. Experiments on the NWPU VHR-10 dataset show that FRIFB surpasses state-of-the-art methods in accuracy and efficiency, making it ideal for large-scale applications.	[155]
	2022	UIU-Net, a novel framework for infrared small object detection	UIU-Net is a novel framework for infrared small object detection, embedding a tiny U-Net within a larger U-Net backbone to enhance multi-scale feature representation. It integrates RM-DS for high-resolution deep supervision and IC-A for improved local contrast. Experiments on SIRST and Synthetic datasets show superior accuracy and generalization, making UIU-Net ideal for real-world applications like video-based infrared object detection.	[154]
	2022	Faster R-CNN	In this study, the authors provide a Faster R-CNN-based model that optimizes loss functions and combines saliency detection, proposal generation, and bounding box regression to enhance detection. The Saliency-Driven Faster RCNN Model for Object Detection and Recognition employing a Computer Vision Approach (SGFr-RCNN) is the name of the suggested technique.	[27]
	2023	Region-based Convolutional Neural Network (R-CNN)	The Region-based Convolutional Neural Network (R-CNN) architecture was used in this work to automatically crop pancreatic computed tomography images.	[21]

Table 1 (continued)

Sr. No.	Year	Algorithm/ Method	Discussion	Refs.
	2023	YOLO v5, v7 and v8	This paper utilizes YOLOv8, a cutting-edge object detection algorithm, to accurately detect tobacco plants. A thorough evaluation and comparison of 12 advanced YOLO object detection algorithms, including YOLOv8, is conducted for tobacco detection.	[50]
	2023	MoViNet	This paper introduces and provides a comprehensive evaluation of two advanced methods for face liveness detection: the Restrictive Voting approach and the MoViNet-based Transfer Learning approach for Robust Face Liveness Detection.	[85]
	2023	LRR-Net	LRR-Net is a deep learning framework for hyperspectral anomaly detection, combining low-rank representation with neural networks for improved accuracy and efficiency. Using ADMM, integrates prior knowledge, reducing manual tuning. An extended version, LRR-Net-S, enhances detection with embedded sparsity. Experiments show that LRR-Net surpasses state-of-the-art methods across multiple hyperspectral datasets.	[152]
	2024	MuDet, a multimodal collaboration network	MuDet is a multimodal collaboration network designed for detecting dense and occluded vehicles in large-scale events, overcoming the limitations of traditional RGB-based methods. It leverages two new multimodal vehicle detection datasets combining RGB imagery with height maps for improved accuracy. MuDet features Uni-Enh for refining modality features, Mul-Lea for cross-modal fusion, and He-Dis for distinguishing complex samples. Experimental results show its superior performance in remote sensing object detection for large-scale disaster scenarios.	[153]

context because they involve two separate processes. A CNN is used to feed images into a feature extractor in a single-stage object detector. The generated features are then used directly for bounding box coordinate regression and classification. Although single-stage object detectors are incredibly efficient and useful for real-time object identification, they can sometimes perform worse than two-stage detectors. Although transformers were first developed for natural language processing, their versatility aided in their transition to visual applications. With their massive hardware resource requirements and severe overparameterization, deep neural networks (DNNs) provide a significant challenge for inference applications, particularly for edge devices with limited resources. Optimizing the overparameterized DNNs and creating lightweight, high-efficiency DNN algorithms are the two main ways to get around this problem. The primary goal of lightweight model design is to build more efficient network computation methods, especially for convolution operations.

4.1. Traditional methods

The majority of traditional object detection methods were created with hand-crafted characteristics. Before 2012, due to the lack of efficient image representation techniques, researchers had no choice but to create highly complex feature vector representations. They also had to implement various speed-up techniques to compensate for the limitations of computational resources [135]. The traditional and deep learning approaches for object detection methods are shown in Fig. 2. There are three phases in traditional object detection methods as given below.

1. Region selection – The objects may appear in various regions of an image due to their varying magnitude and aspect ratio. Therefore, specifying an object's region is crucial from the first step. Consequently, a multi-scale sliding window technique is used to examine a complete image to identify objects. However, this method results in a lot of irrelevant choices and has a huge computational cost.
2. Feature extraction – Following object location, a feature extraction procedure is performed to generate a reliable representation.
3. Classification – At this phase, a classifier is employed to recognize the target items and construct more structured and significant models for the perception of images.

In this section, Viola Jones, Histogram of Oriented Gradients (HOG), Deformable Parts Model (DPM), Scale-Invariant Feature Transform (SIFT), Speeded-Up Robust Features (SURF), Oriented FAST and Rotated BRIEF (ORB), Gray Level Co-occurrence Matrix (GLCM) traditional methods are explained in detail and the architectures of the same are illustrated in Fig. 3.

4.1.1. Viola Jones

In their 2001 publication "Rapid object detection using a Boosted Cascade of simple features," Paul Viola and Michael Jones presented the Viola-Jones algorithm, a machine-learning method for object detection. The algorithm was primarily conceived for face detection. Despite it has lower accuracy than modern face detection methods based on Convolutional Neural Networks (CNNs), the Viola-Jones algorithm is still an efficient solution for resource-constrained devices. Jing Huang et al. [1] suggest a better Viola-Jones face detection algorithm based on Hol-oLens, which improves upon the traditional Viola-Jones face detection algorithm that uses Haar-like rectangular feature expansion to increase detection speed and efficiency. This algorithm is based on two-dimensional convolution separation and image resampling.

In the Viola-Jones algorithm, given a grayscale image, the algorithm

analyses many windows of different sizes and positions and tries to detect the target object by looking for specific image features in each window. The Viola-Jones algorithm is based on Haar-like features, integral images for accelerating the feature computation, AdaBoost learning for feature selection and the cascade of classifiers for fast rejection of windows without faces [2]. The algorithm uses a set of features like Haar wavelets, which are a set of square-shaped functions. Each feature defines a group of rectangles within an image window, where each rectangle is defined and as either white or black. The feature value is calculated by finding the difference between the sum of pixel values in the white rectangles and the sum of pixel values in the black rectangles. In 'flat regions,' where all pixels have similar values, the feature value will be close to zero. Conversely, a large feature value occurs in regions where there is a significant difference between the pixel values in the black and white rectangles. The integral image is a data structure for efficiently computing the sum of pixel values in a rectangular image window. Hence, the Haar-like features can be computed very quickly using the integral image representation. The Viola-Jones algorithm uses AdaBoost to find the best features and to train a classifier. Each Haar-like feature represents a weak classifier. The final classifier is given by a linear combination of weak classifiers. Larger weights are associated with better classifiers using the AdaBoost learning algorithm.

4.1.2. Histogram of Oriented Gradients (HOG) detector

In computer vision, many algorithms are designed to extract spatial features to identify objects using information about image gradients. Seemanthini K et al. [3] propose a novel approach to human object detection based on cluster segmentation where each frame undergoes cluster segmentation, followed by feature extraction using the Histogram of Oriented Gradients (HOG) and finally, classification is performed using the Support Vector Machine (SVM) algorithm. A histogram is an approximate representation of the distribution of numerical data that looks like a bar graph. Each bar represents a group of data that falls in a certain range of values, called bins. HOG will produce a histogram of gradient directions in an image, and it is a feature descriptor like the Canny edge detector and SIFT. S. Bougharriou et al. [4] propose a method for vehicle detection, which includes HOG and SVM for feature extraction and classification respectively.

In the HOG detector, the image is first converted to grayscale, and the gradient at each pixel is computed. This involves calculating the gradients in both the x and y directions using filters like the Sobel operator. Each pixel's gradient vector is then used to calculate its magnitude and orientation. These values are grouped into bins based on their orientation. Typically, orientations are divided into bins of 0 to

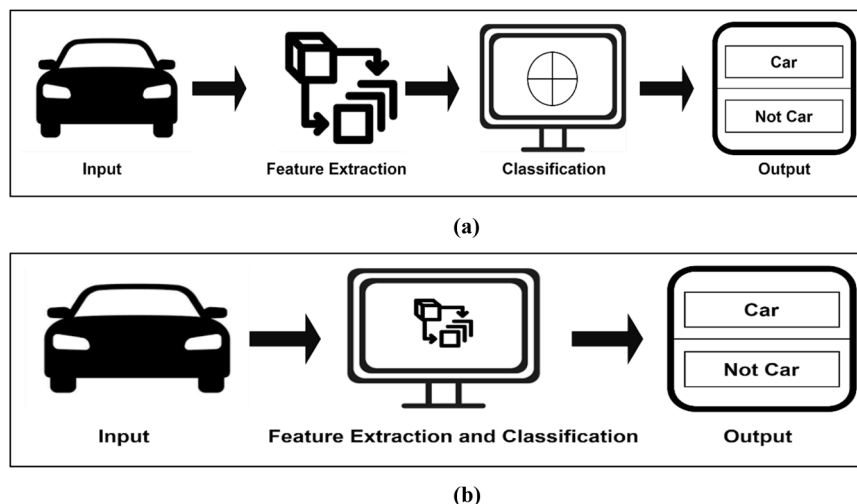
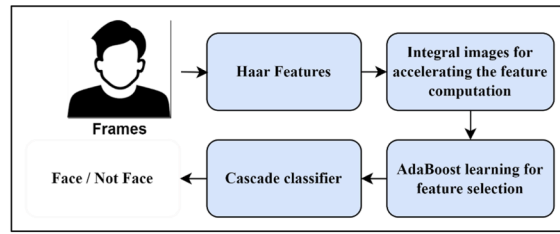
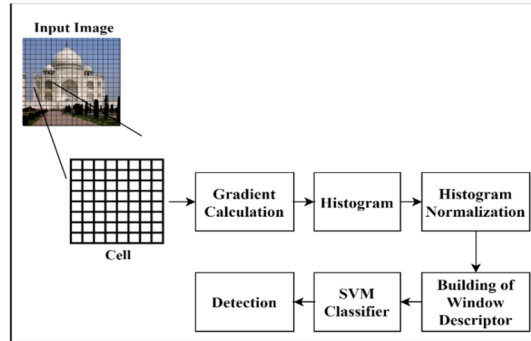


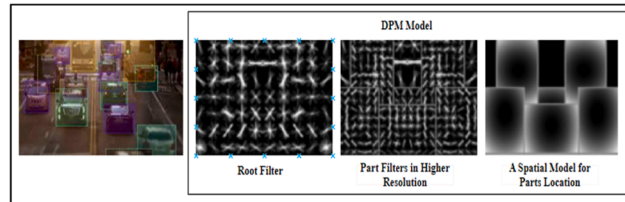
Fig. 2. Object detection methods (a) Traditional approach (b) Deep learning approach.



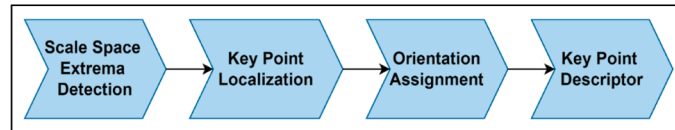
(a)



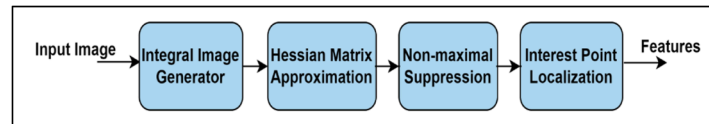
(b)



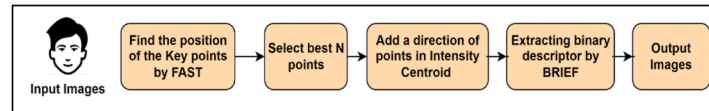
(c)



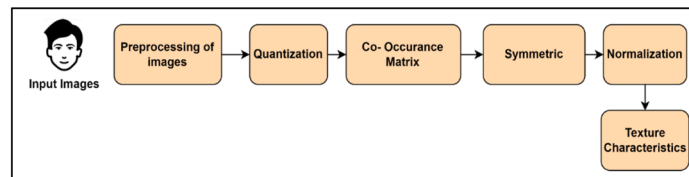
(d)



(e)



(f)



(g)

Fig. 3. Architectures of traditional methods (a) Viola Jones (b) HOG (c) DPM (d)SIFT (e) SURF (f) ORB (g) GLCM.

180° (for unsigned gradients) or 0 to 360° (for signed gradients). The image is divided into small, connected regions called cells, typically 8×8 pixels in size. A histogram of gradient directions is computed for the pixels within each cell, with the contribution of each pixel weighted by the gradient magnitude. To make the descriptor more robust to changes in illumination and contrast, several adjacent cells are grouped into larger blocks. The histograms within each block are normalized, which reduces the effect of local variations in lighting and contrast. The normalized histograms from all blocks are concatenated to form a single feature vector. This feature vector can then be used as an input to machine learning algorithms for classification tasks. HOG detectors can be used for different detection applications along with suitable classifiers.

4.1.3. Deformable Parts Model (DPM)

The Deformable Parts Model (DPM) is a classical object detection framework that gained popularity due to its flexibility in modelling objects with varying shapes and articulations. The HOG detector models an object class as a single rigid template. But objects are composed of deformable parts. The DPM model starts by borrowing the idea of the HOG detector. It takes a HOG template for the full object. The DPM recognizes objects with a mixture of graphical models (Markov random fields) of deformable parts. K. Wang et al. [5] propose a novel model named Convolutional Deformable Part Model (CDPM) by enhancing a traditional DPM. CDPM simplifies the complex process of part alignment by breaking it into two manageable steps: first, a vertical alignment step identifies each body part along the vertical axis, and second, a horizontal refinement step, guided by attention mechanisms, reduces background noise surrounding the detected body parts. The basic DPM consists of three main components: the root filter, part filters, and a spatial model. The root filter defines a detection window that roughly covers the entire object. A filter specifies the weights of a region's feature vector. Multiple-part filters, which are learned at twice the resolution of the root filter, cover smaller sections of the object. The spatial model is used to score the relative positions of the part filters concerning the root filter [7].

DPM allows for object parts to move independently, making it suitable for detecting objects that exhibit variations in pose, deformation, or viewpoint. Each part of an object can be represented separately, which provides flexibility in capturing complex object shapes. Since DPM detects parts of an object independently, it can better handle partial occlusion. If some parts of an object are occluded, DPM can still detect the object based on the visible parts. It performs well in cluttered environments, where not all parts of the object are visible. J. Li [6] presents a framework for multiple object detection, which includes the R-CNN and the DPM. To further refine the candidate recommendations produced by the DPM, they also suggested a new filter relying on a dense subgraph discovery technique.

In scenarios where labelled data is limited, DPM can perform relatively well compared to deep learning models that require large datasets for effective learning. While DPM has been surpassed in accuracy and speed by modern deep learning-based object detection methods like Faster R-CNN and YOLO, its ability to handle deformations and occlusion in a part-based manner remains a key advantage.

4.1.4. Scale-Invariant Feature Transform (SIFT)

Scale-Invariant Feature Transform, or SIFT, was introduced by D. Lowe of the University of British Columbia in 2004. SIFT is invariant to the rotation and scale of the image. The SIFT method consists of four phases, the first of which is the scale-space peak selection, which indicates a possible feature placement. Keypoint localization is the second phase, which involves precisely identifying the feature key points; orientation assignment is the third step, which entails giving key points an orientation; and keypoint description and matching are the final steps [8]. SIFT's advantages include local features that are robust against clutter and occlusion, the ability to match individual features to a large database of objects, the ability to generate numerous features for even

small objects, and the ease with which a variety of feature types can be added, each of which increases robustness [10]. Liang Zheng et al. [9] provide a comprehensive survey of image retrieval methods where they classified various methods into different categories. Out of that one is SIFT SIFT-based category due to its advantages in dealing with image transformations.

4.1.5. Speeded-Up Robust Features (SURF)

The SURF method is a fast and robust approach for local, similarity-invariant image comparison and representation. The main attraction of the SURF technique is its speedy computation of operators through the use of box filters, which enable real-time applications such as tracking and object detection. SURF is composed of two steps, feature extraction and feature description. SURF is robust to changes in scale and rotation, meaning it can detect the same object or feature even if the object appears larger, smaller, or rotated in the image [12]. This invariance is essential in object recognition tasks where objects may appear in different orientations. SURF performs well with blurred and rotated images, but poorly with changing viewpoints and lighting conditions. Madhumita P. et al. [11] propose image hashing based on shapecontext via Speeded Up Robust Features (SURF). In their work, they concluded that the SURF-based method is more resilient to many kinds of image modifications than SIFT and is several times faster than SIFT.

4.1.6. Oriented FAST and Rotated BRIEF (ORB)

Oriented FAST and Rotated BRIEF (ORB) is a feature detection and descriptor algorithm designed to be fast and efficient while maintaining accuracy. ORB combines two techniques: FAST (Features from Accelerated Segment Test) for keypoint detection and BRIEF (Binary Robust Independent Elementary Features) for feature description, with added enhancements for orientation and rotation invariance. It integrates the FAST keypoint detection method with the BRIEF descriptor while adding orientation and rotation invariance to the mix. This algorithm is rotation invariant and resistant to noise. The BRIEF descriptor is a binary descriptor. This makes ORB extremely fast to compute and match compared to floating-point descriptors like those in SIFT and SURF [14]. Binary descriptors allow for efficient matching using the Hamming distance, making ORB much faster than traditional approaches, especially in low-resource environments. ORB's keypoint detection, which is based on the FAST algorithm, provides robustness against image noise and minor blurring. This makes it highly effective for detecting features in images that are imperfect or of low quality. This robustness is particularly useful in practical applications such as surveillance, mobile vision, or navigation in complex environments. Helia Sharif and Matthew Holzel [13] explore the impact of prefiltering on object detection using Oriented Fast and Rotated BRIEF (ORB) techniques. ORB relies on the minimum distance between matched feature descriptors to confirm the presence of an object, highlighting the trade-off between execution time and the minimum Hamming distance during feature matching.

4.1.7. Gray Level Co-Occurrence Matrix (GLCM)

In image processing, the Gray Level Co-occurrence Matrix (GLCM) is a widely used technique for texture analysis. It determines the frequency of pixel pairings in an image that has particular values and spatial connections. Texture-describing measurements of statistics are frequently extracted using the GLCM. It considers the relationships between neighbouring pixels, providing rich texture information. This makes GLCM particularly well-suited for tasks that require analysing fine-grained texture patterns, such as medical image analysis. The key concept of GLCM lies in analysing how often grey levels occur together within an image, specifically considering neighbouring pixels. By counting co-occurrences in a specific pixel offset and specific directions, GLCM captures the spatial arrangement of textures [16].

The Distance (d) means displacement between two pixels, Angle (θ) the direction in which pixel pairs are considered, typically in 0° , 45° ,

90°, and 135° and the Number of Gray Levels (G) means the number of discrete intensity levels in the image parameters used to calculate GLCM. Once the GLCM is computed, various statistical measures can be derived from it to characterize the texture and structure of the image. Contrast, Homogeneity, Dissimilarity, Correlation, and Energy are some common features derived in GLCM. GLCM can be calculated in different orientations, making it effective for rotation-invariant texture analysis. It is applicable across domains like satellite imagery, agriculture, and medical diagnostics for segmentation, classification, and analysis. Priyanka and Dharmender Kumar [15] propose feature extraction and selection of kidney Ultrasound Images using GLCM and PCA. The proposed method generates several second-order statistical texture features using the Gray Level Co-occurrence Matrix (GLCM), including energy, entropy, homogeneity, correlation, contrast, and dissimilarity.

4.2. Deep learning models

Deep learning-based object detection methods are broadly categorized into four types: two-stage detectors, one-stage detectors, transformer-based detectors, and lightweight networks. A small collection of region proposals is first generated using two-stage detectors, which then extract feature vectors from these proposals, encode them using deep convolutional neural networks, and utilize them to forecast predetermined classes [136]. One-stage detectors, on the other hand, try to categorize each region of interest as either a target object or background by treating every place in the input image as a possible target object. Although one-stage detectors are quicker and better suited for applications requiring real-time detection of objects, they generally demonstrate lower performance compared to two-stage detectors, which often achieve superior results on many publicly accessible datasets [110]. The architecture of the one-stage and two-stage detectors is

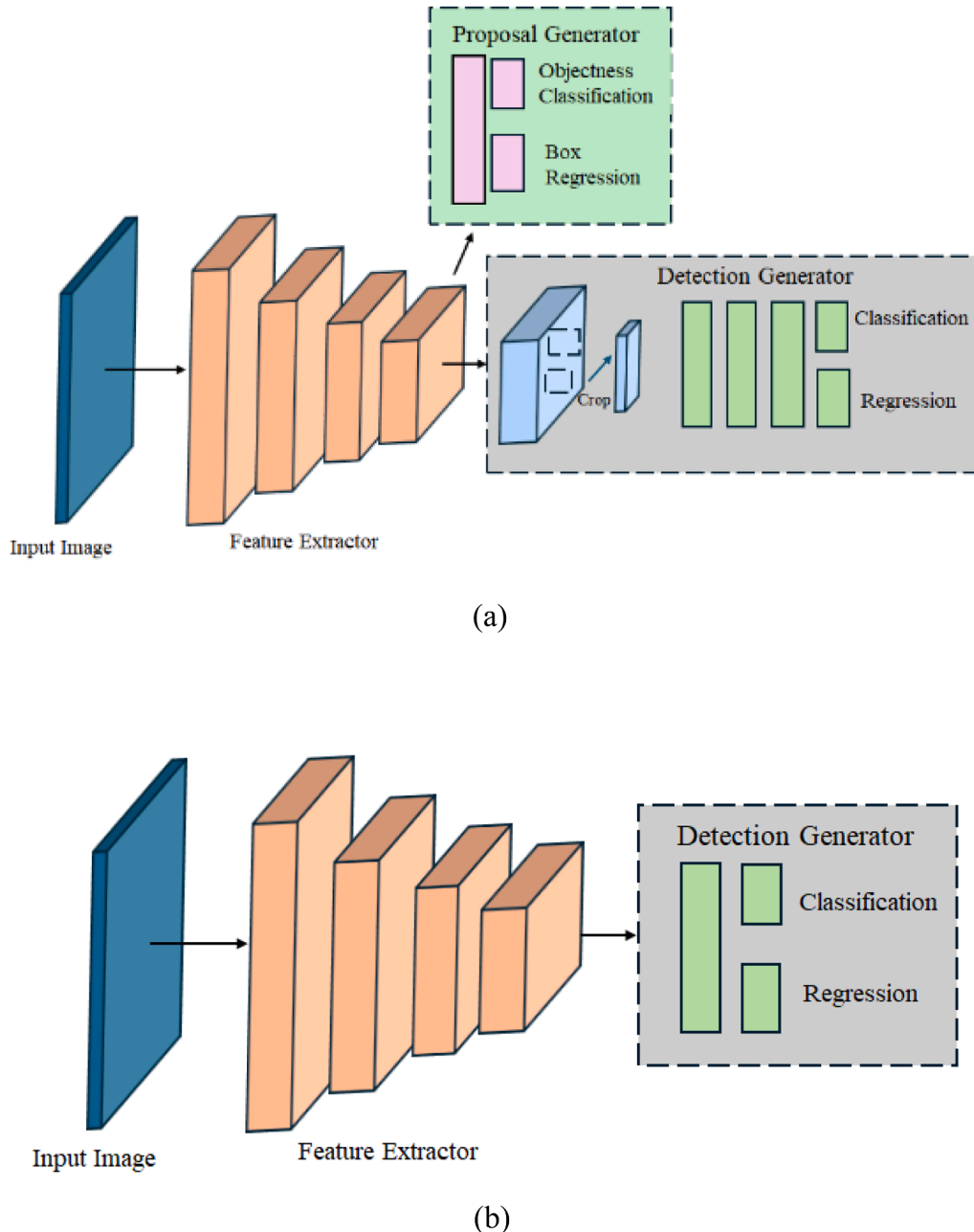


Fig. 4. Types of detectors (a) Two-stage detectors (b) One-stage detector.

shown in Fig. 4.

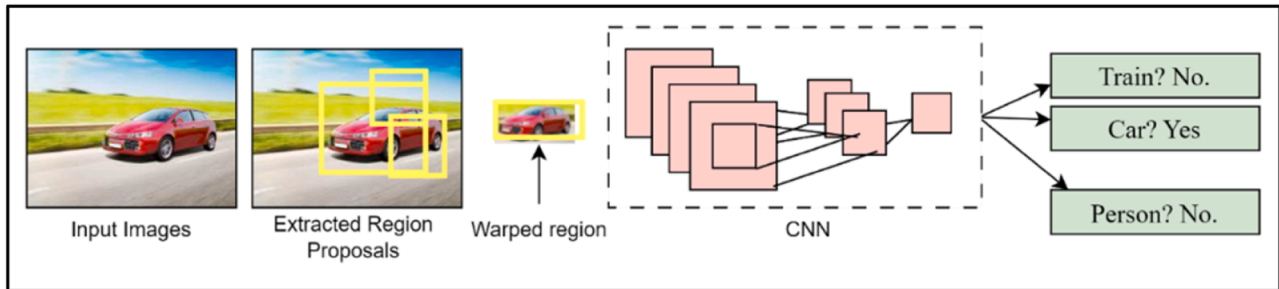
4.2.1. Two stage detectors

The two-stage object detector divides the whole process into 2 steps:

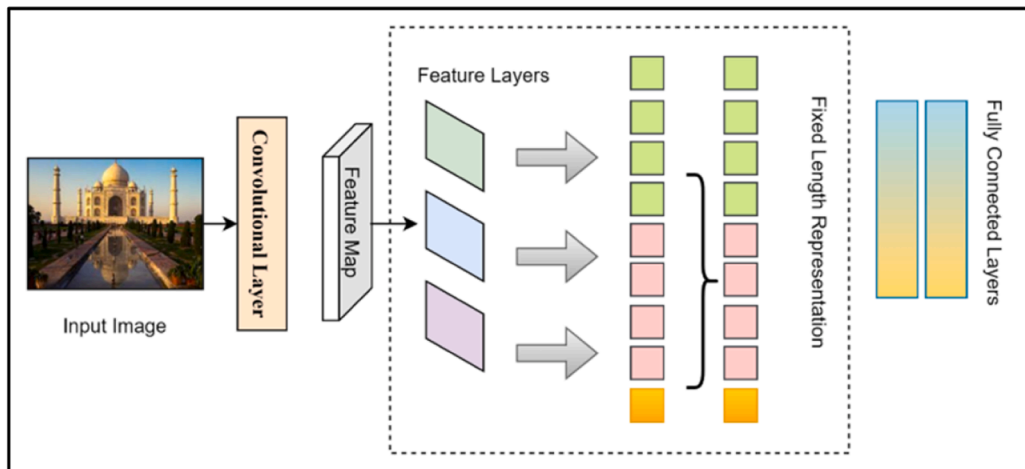
Step 1: It first extracts the features using a CNN

Step 2: It then extracts a series of regions of interest called object proposals and then the classification and localization happen only on the object proposals.

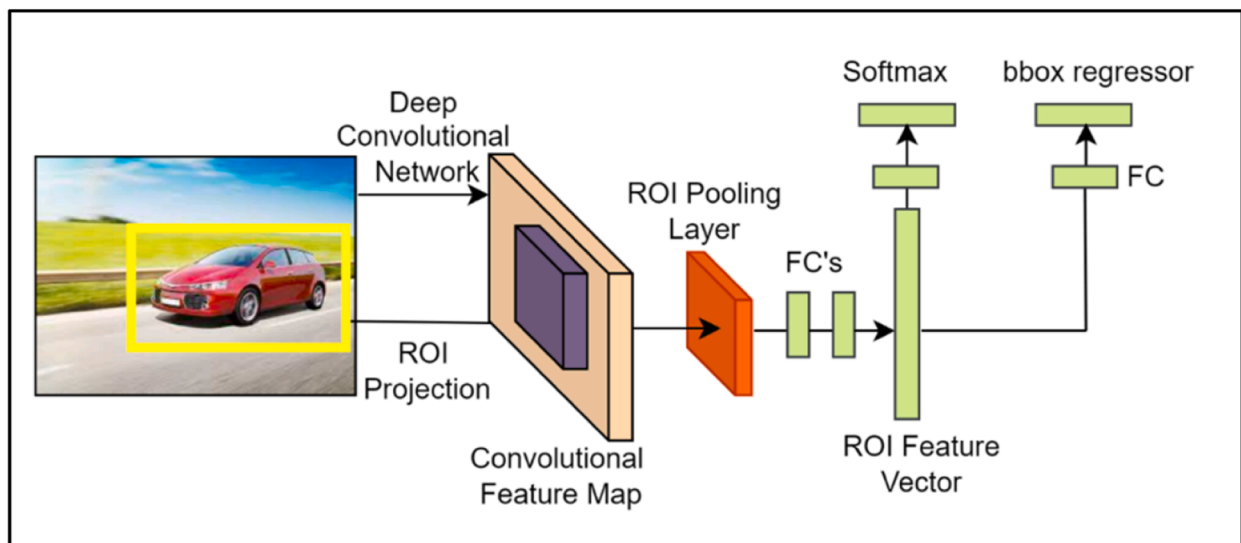
Two-stage object detectors are very powerful and extremely accurate having very high values of mAP [109]. Hence, they are mostly used in



(a)

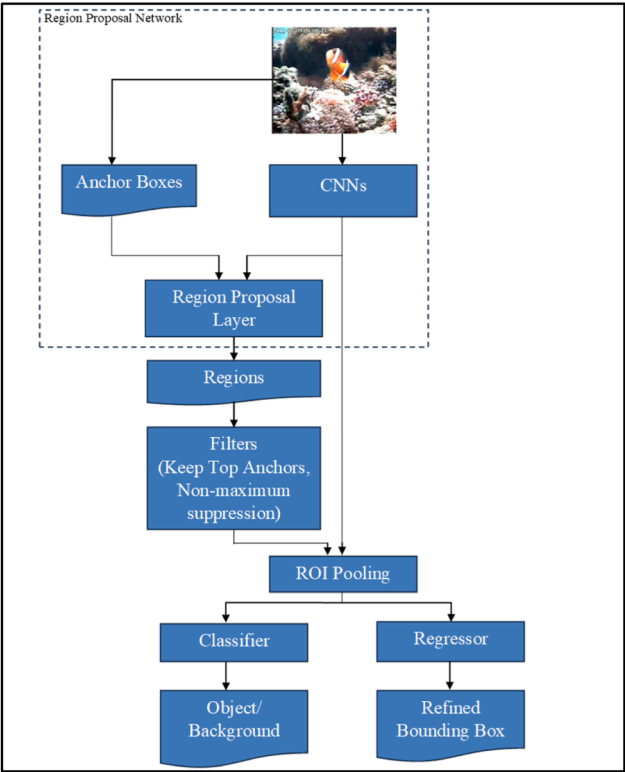


(b)

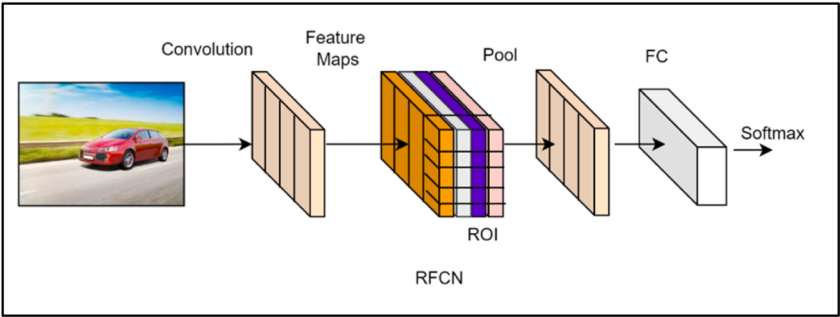


(c)

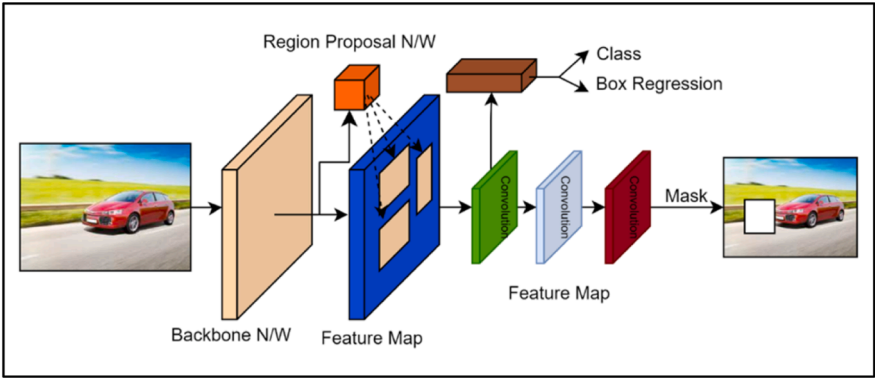
Fig. 5. Two-stage detectors architecture (a)RCNN (b)SPP-Net (c) Fast R-CNN (d) Faster R-CNN (e) R-FCN (f) Mask R-CNN.



(d)



(e)



(f)

Fig. 5. (continued).

the medical domain where classification accuracy is more important than speed. Fig. 4. (a) illustrates the two-stage object detectors. Here, Region-based Convolutional Neural Network (R-CNN), Spatial Pyramid Pooling (SPP-Net), Fast R-CNN, Faster R-CNN, R-FCN, Mask R-CNN, DetectorRS are explained in detail [136]. The architecture of the two-stage detectors is presented in Fig. 5.

4.2.1.1. Region-based Convolutional Neural Network (R-CNN). R-CNN, introduced by Ross Girshick in 2014, is a pioneering object detection framework that follows a three-step process. First, it generates Regions of Interest (ROIs) using methods like selective search or edge boxes, which identify potential objects of interest within the image through bounding boxes. Next, a Convolutional Neural Network (CNN) extracts features from each ROI, representing them in a compact and meaningful way [20,136]. Finally, these features are fed into a classifier, such as a Support Vector Machine (SVM), to determine the object class within each ROI. A key advantage of R-CNN is its ability to handle multiple object classes effectively, as the classifier is trained separately for each class. However, its main drawback is its computational inefficiency, as the CNN must process each ROI individually, making it resource-intensive and slow. Mamta Juneja et al. [21] proposed R-CNN for auto-cropping of pancreatic computed tomography. They implemented R-CNN to localize the pancreas in the pancreatic CT scans.

4.2.1.2. Spatial Pyramid Pooling (SPP-Net). SPP-Net is a convolutional neural network architecture that utilizes spatial pyramid pooling to eliminate the network's fixed-size input constraint. A Spatial Pyramid Pooling (SPP) layer is integrated on top of the final convolutional layer, where it pools features and generates fixed-length outputs. These outputs are then passed to the fully connected layers. By aggregating information at a deeper stage in the network hierarchy, between the convolutional and fully connected layers, SPP-Net eliminates the need for cropping or warping input images at the beginning of the network. Spatial Pyramid Pooling (SPP) is used to extract features at different sizes of regions. In SPP-Net, the image is passed through the CNN only once, and region features are extracted from the fifth convolutional layer. The workflow of SPP-Net begins with the input image, followed by the convolutional layers, then the spatial pyramid pooling layer, and finally the fully connected layers to produce the output. This approach addresses the drawback of R-CNN which suffers from slow performance due to extensive calculations [22]. The parameters are shared by CNN layers, so this network is fast as compared to the R-CNN. The advantage of SPP-Net is its ability to adapt to varying input sizes, as the input size limitation is determined by the fully connected layers, not the convolutional layers. In contrast, R-CNN requires cropping and warping the input to a fixed size. The Spatial Pyramid Pooling layer in SPP-Net eliminates this size constraint, allowing the network to process inputs of different dimensions without the need for resizing. Anita Mathew and Sony Kurian [23] applied the SPP-Net model for the identification of Malicious code variants.

4.2.1.3. Fast R-CNN. Fast R-CNN, introduced by Ross Girshick in 2015, improves upon R-CNN by addressing its computational inefficiency. Instead of running the CNN on each Region of Interest (ROI) individually, Fast R-CNN processes the entire image with a single CNN and generates ROIs using methods like selective search or edge boxes. The extracted features are then pooled using ROI pooling to create fixed-length feature vectors for each ROI. These vectors are passed through two fully connected layers: one for classification and another for bounding box regression. While Fast R-CNN significantly reduces computational cost compared to R-CNN, it still requires a separate classifier for each object class, which can become costly for large numbers of classes [24,136]. Unlike R-CNN, which processes each region proposal independently, Fast R-CNN computes the CNN feature map once for the entire image and reuses it for all region proposals. This

reduces the computational overhead and makes it faster. Fast R-CNN uses a Region of Interest (RoI) pooling layer to extract fixed-size features from variable-sized regions, leading to better alignment of features with proposed regions, and enhancing detection accuracy. In R-CNN, each region proposal is passed through a CNN individually, requiring storage of multiple CNN outputs. Fast R-CNN avoids this by processing the entire image once, significantly reducing memory and storage needs [25].

4.2.1.4. Faster R-CNN. Faster R-CNN, introduced by Shaoqing Ren et al. in 2015, enhances Fast R-CNN by further reducing computational costs. It achieves this by using a single CNN to both generate ROIs and extract features, instead of using separate networks for each task as in Fast R-CNN. A fully convolutional network (FCN) generates the ROIs from the CNN-extracted features [136]. These features are then pooled into fixed-length vectors using ROI pooling and passed through two fully connected layers: one for classification and one for bounding box regression. Faster R-CNN improves efficiency and streamlines the object detection process. W. Wu et al. [26] propose face detection with different scales based on faster R-CNN. Faster R-CNN combines the feature extraction and region proposal steps from both R-CNN and Fast R-CNN into a single CNN, significantly enhancing computational efficiency. Additionally, it incorporates an anchor box mechanism to manage multiple scales and aspect ratios, improving the robustness of object detection [86]. The key difference between R-CNN, Fast R-CNN, and Faster R-CNN lies in their approach to generating region proposals. R-CNN relies on selective search, Fast R-CNN uses RoI pooling, and Faster R-CNN employs a Region Proposal Network (RPN) to directly generate region proposals from the CNN feature maps. With the right data and some programming expertise, these methods can be used to accurately detect and classify objects in images and videos [27].

4.2.1.5. R-FCN. Region-based Fully Convolutional Network (R-FCN) is a region-based object detector that is fully convolutional, sharing most of the computation across the entire image [136]. It uses position-sensitive score maps to balance translation invariance in image classification with translation variance in object detection. This approach leads to faster inference times compared to Faster R-CNN while still achieving competitive accuracy. Unlike traditional region proposal networks (RPN) used in R-CNN, Fast R-CNN, and Faster R-CNN, R-FCN removes the fully connected (FC) layers after ROI pooling [29]. Instead, it generates score maps before ROI pooling, and all-region proposals use the same score maps for average voting, which is computationally efficient. This design makes R-FCN faster than Faster R-CNN with similar mean Average Precision (mAP) performance. Jiaxin Si et al. [28] propose a hand-raising gesture detection method in real classrooms using R-FCN.

4.2.1.6. Mask R-CNN. A further development of the Faster R-CNN architecture, Mask R-CNN (Mask Region-based Convolutional Neural Network) is intended to carry out instance segmentation [136]. In addition to identifying objects within an image, this process entails creating accurate segmentation masks for every instance of an object. By including an extra branch for mask prediction, Mask R-CNN extends on Faster R-CNN, which was first presented by Kaiming He et al. in 2017. While its second stage processes in parallel to predict object classes, build bounding boxes, and produce binary masks for each Region of Interest (RoI), the architecture keeps the region proposal generating stage from Faster R-CNN. The Backbone Network, Region Proposal Network (RPN), Mask Representation, and ROI alignment are significant components of Mask R-CNN [30]. One of the main advantages of Mask R-CNN is its ability to deliver accurate pixel-level segmentation masks while efficiently handling multiple tasks within a single framework. The RPN enables the model to focus on relevant regions, thereby reducing computational costs compared to exhaustive scanning methods. Consequently, Mask R-CNN typically requires substantial computing

resources, such as GPUs, to achieve optimal performance Mask R-CNN's capacity to generate precise pixel-level segmentation masks while effectively managing several jobs inside a single framework is one of its key features. Compared to thorough scanning techniques, the RPN lowers computational costs by allowing the model to concentrate on pertinent regions. Nevertheless, the total computing load rises with the addition of the mask prediction branch. As a result, Mask R-CNN usually needs a lot of processing power, including GPUs, to perform at its best [31].

4.2.1.7. DetectorRS. The "look and think twice" method is utilized by several contemporary two-stage detectors, in which object proposals are initially created before features are extracted for object recognition. This technique is used by DetectorRS at both the network's macro and micro levels. The Recursive Feature Pyramid (RFP), which stacks several Feature Pyramid Networks (FPNs) with extra feedback connections from the top-down path of the FPN to the bottom-up layers, is introduced at the macro level. Before entering the subsequent FPN layer, the output is defined using a Atrous Spatial Pyramid Pooling (ASPP) layer. Using an attention map, a Fusion module aggregates FPN outputs from many modules [141]. An average pooling layer with a 5×5 filter and a 1×1 convolution acts as a switch function to regulate the dilation rate, enabling the network to detect objects at various scales. Switchable Atrous Convolution (SAC) is also placed between two global context modules to stabilize the switching process. The combination of these techniques, Recursive Feature Pyramid and Switchable Atrous Convolution, forms DetectorRS. S. Qiao et al. [32] propose a model based on

DetectorRS for detecting objects with recursive feature pyramid and SAC.

The characteristics and observation remarks of the two stage models are given in Table 2.

4.2.2. One stage detectors

Single-stage detectors perform both classification and localization of semantic objects in a single pass through dense sampling [107]. The architecture of the stage detectors is illustrated in Fig. 4(b). They utilize predefined boxes of different scales and aspect ratios to localize objects. This gives them an advantage over two-stage detectors in terms of real-time performance and simpler design [108]. Here, DetectorNet, OverFeat, You Only Look Once (YOLO), Single Shot Detector (SSD), YOLOv2, RetinaNet, YOLOv3, CenterNet, EfficientDet, YOLOv4, YOLOv5 and YOLOv7 are explained in detail.

4.2.2.1. DetectorNet. DetectorNet is a framework developed by Ross Girshick and others that focuses on addressing limitations in traditional object detection methods by introducing a model that directly detects objects from a 3D bounding box perspective. It is primarily geared toward handling tasks like object pose estimation and occlusion detection, where traditional 2D detection methods may struggle. DetectorNet employs a region-based approach for object detection, where it first identifies regions of interest (ROIs) and then performs 3D bounding box estimation to localize objects more accurately [142]. Instead of relying solely on 2D bounding boxes, DetectorNet leverages a deeper understanding of the object's orientation and depth in the scene. It uses CNNs

Table 2
Characteristics of two-stage models.

Sr. No.	Method	Year	Image Input Size	Region Proposal Method	Loss/ Cost Function	Remarks	Refs.
1.	R-CNN	2014	Variable usually cropped to 227×227 or 224×224	Selective Search	Softmax loss for classification, Smooth L1 loss for bounding box regression	R-CNN introduced CNNs for object detection, significantly improving accuracy over traditional methods and achieving high accuracy with fine-tuned models. However, it is very slow due to separate steps for region proposal, feature extraction, and classification, with high computational cost and storage requirements.	[102]
2.	SPP-Net	2014	Variable (due to spatial pyramid pooling layer)	Selective Search	Softmax loss for classification, Smooth L1 loss for bounding box regression	SPP-Net introduced spatial pyramid pooling for fixed-size feature maps, allowing variable input sizes and improving speed over R-CNN through shared convolutional computation. However, it still relies on Selective Search, making it slower than later models, and lacks end-to-end training.	[103]
3.	Fast R-CNN	2015	Variable typically resized to 224×224	Selective Search	Multi-task loss (classification + bounding box regression)	Fast R-CNN is faster than R-CNN with shared convolutional layers, enabling end-to-end training and higher accuracy. However, it still relies on Selective Search for region proposals, which limits its speed.	[102]
4.	Faster R-CNN	2015	Variable, typically 600×1000	Region Proposal Network (RPN)	Multi-task loss (classification, bounding box regression, RPN loss)	Faster R-CNN offers fully end-to-end training with an integrated RPN for faster region proposals, achieving high accuracy and efficiency in object detection. However, it is slower than single-shot detectors like YOLO or SSD and is computationally intensive.	[102]
5.	R-FCN	2016	Variable	Region Proposal Network (RPN)	Softmax loss for classification, Smooth L1 loss for bounding box regression	R-FCN's fully convolutional architecture makes it faster than Faster R-CNN, offering good accuracy with reduced computational complexity. However, it is less accurate for small objects and has limited flexibility for adapting to specific tasks.	[104]
6.	Mask R-CNN	2017	Variable, typically 800×800	Region Proposal Network (RPN)	Multi-task loss (classification, bounding box regression, mask prediction)	Mask R-CNN extends Faster R-CNN with a pixel-level segmentation branch, excelling in instance segmentation and detection tasks. However, it requires higher computational resources and has a slower inference speed compared to Faster R-CNN.	[105]
7.	DetectorRS	2021	Variable, typically 640×640 or 1024×1024	Transformer-based feature extraction with anchor-free mechanisms	Focal loss for classification, IoU-based loss for bounding boxes	DetectorRS combines transformer-based architectures with object detection, offering high accuracy and scalability for large datasets. However, it is computationally expensive, especially during training, and as a newer model, it has limited community adoption and fewer pre-trained resources.	[106]

to extract features from images, followed by a component that attempts to predict the spatial configuration of the objects. The DetectorNet is enhanced by Transformer. To create correct predictions, it has two modules: Multi-view Temporal Attention and Dynamic Attention. By constantly updating the acquired knowledge, these modules focus on dynamic spatial correlation and long-distance and short-distance temporal correlation, respectively [143]. The advantage of DetectorNet is that it can estimate the 3D pose and bounding box of objects, making it especially useful in scenarios that require spatial understanding, such as autonomous driving, robotics, and augmented reality. One of its strengths is detecting objects that are partially occluded. Traditional 2D object detection methods often fail when parts of the object are hidden, but DetectorNet can use its 3D estimation to infer the presence of the full object. By using 3D bounding boxes, DetectorNet provides more accurate object localization than traditional 2D methods, especially in cluttered or crowded environments where overlapping objects are present [144]. At the same time, the integration of 3D bounding boxes and object pose estimation introduces additional complexity in both the training and inference stages, making the model more computationally expensive compared to 2D detection methods like Fast R-CNN or YOLO. DetectorNet requires 3D annotations and pose data for training, which are more difficult and expensive to obtain compared to 2D bounding boxes. Annotating 3D objects is more involved and time-consuming [145].

4.2.2.2. OverFeat. One of the first models to combine object identification, localization, and classification into a single convolutional neural network is OverFeat. The primary concept is to classify images at various points on regions of various picture scales using a sliding window technique, and then use a regressor trained on top of the same convolution layers to forecast the locations of the bounding boxes. It uses the image classification job to train a CNN model [137]. Next, train a regression network to predict object bounding boxes at every spatial location and scale by substituting it for the top classifier layers. Each regressor is created for a single image class and is class-specific. Images with a classification and a bounding box are the model's input. The four values that indicate the coordinates of the bounding box edges x_{left} , x_{right} , y_{top} , and y_{bottom} are the model's output. For every training example, the regressor is trained to minimize the l2 norm between the generated bounding box and the ground truth. During detection, it uses a pre-trained CNN model and predicted object bounding boxes to classify each region identified by the classifier. The bounding boxes are then merged based on sufficient overlap in localization and high confidence from the classifier, ensuring they correspond to the same object. The advantage of the OverFeat algorithm is that it integrates classification, localization, and detection tasks in a single architecture [33]. This multitask learning reduces redundancy in computations and improves overall efficiency, also performs detection by running the convolutional network at multiple scales, enabling effective detection of objects of varying sizes in an image. Y. Saleh and E. Edirisinghe [34] propose a system for 3D face reconstruction and recognition using the OverFeat network.

4.2.2.3. You Only Look Once (YOLO). YOLO (You Only Look Once) is a real-time object detection system that processes an image in a single pass [137]. It divides the input image into an $S \times S$ times $S \times S$ grid (e.g., $S = 7$) and assigns each grid cell the responsibility for detecting objects whose centres fall within it. Each cell predicts B bounding boxes, along with associated confidence scores and conditional class probabilities. The bounding boxes include coordinates (x, y) , dimensions (w, h) , and confidence, which reflect the Intersection over Union (IoU) with ground truth boxes [35]. YOLO's architecture uses convolutional layers and fully connected layers, framing detection as a single regression problem, which enhances speed and simplicity. While YOLO excels in real-time applications like autonomous driving and surveillance, it struggles

with detecting small or densely packed objects due to its fixed grid approach. Liu W. et al. [36] proposed an image-adaptive YOLO for object detection in adverse weather conditions.

4.2.2.4. Single Shot Detector (SSD). The Single Shot Detector (SSD) is intended for real-time object detection. The SSD object detection process consists of two steps: first, feature maps are extracted, and then convolution filters are applied to identify objects. A pre-trained base network, such as VGG or ResNet, that has been trained on large-scale image classification dataset, like ImageNet, makes up the architecture of an SSD. Some other layers, also referred to as "extra layers," are then placed on top of this base network. These additional layers, which are usually made up of convolutional and pooling layers, are in charge of detecting objects at many different scales [37]. SSDs' speed and efficiency are their primary advantages. Their ability to detect objects in real time due to their usage of a single network makes them appropriate for applications like surveillance systems and self-driving cars. Additionally, SSDs can benefit from a vast amount of labelled data accessible for image classification jobs because they employ a pre-trained base network. This enables them to learn on relatively limited datasets and still achieve excellent accuracy [137]. SSDs have the drawback of being less precise than other techniques, including the R-CNN family of techniques. This is because SSDs are unable to benefit from the extra context and data that numerous networks offer when they are only connected to one network. Second, the scale of the things in an image may affect them. SSDs may have trouble seeing objects that are much smaller or larger than the objects in the training dataset because the additional layers are made to recognize objects at different scales. Zhou, S. and Qiu, J. [38] propose an improved SSD for object detection that has interactive multi-scale attention capabilities.

4.2.2.5. YOLOv2. YOLOv2 introduces several improvements over YOLOv1. It incorporates Batch Normalization (BN) across all convolutional layers, improving the mean Average Precision (mAP) by 2 %. YOLOv2 eliminates all fully connected layers and adopts anchor boxes for bounding box prediction. To enhance output resolution, one pooling layer is removed, and the detection network is trained on 416×416 images, producing a 13×13 feature map output ($32 \times$ downsampled) [131]. The anchor box sizes and scales are pre-defined through k-means clustering without prior information, unlike Faster R-CNN. Additionally, YOLOv2 stabilizes location prediction by using logistic activation (σ), ensuring that location values are constrained between 0 and 1 [132]. While the 13×13 feature map is sufficient for detecting large objects, small objects are detected by mapping the $26 \times 26 \times 512$ feature maps from earlier layers into a $13 \times 13 \times 2048$ feature map, which is then concatenated with the original 13×13 feature maps for improved detection. The advantage of the YOLOv2 is that this network runs faster at smaller-size images [39]. For low-resolution YOLOv2, at 90 FPS, mAP is as good as Fast R-CNN, which is ideal for smaller GPUs. For high-resolution YOLOv2, 76.8 % mAP is obtained at real-time speed. Besides the above modification, network architecture is also a factor affecting the mAP. The darknet-19 classification network is used in YOLOv2 for feature extraction. Darknet-19 has many 1×1 convolutions to reduce the number of parameters and can obtain a good balance between accuracy and model complexity. Darknet-19 has lower model complexity (FLOP) and therefore much faster detection speed. Boudjit, K. and Ramzan, N. [40] propose human detection based on deep learning YOLO-v2 for real-time UAV applications.

4.2.2.6. RetinaNet. RetinaNet is a one-stage object detection model known for achieving high accuracy by using the Focal Loss function. It tackles the problem of class imbalance between background and foreground classes during training. To concentrate learning on concrete negative examples, focal loss adds a modulating term to the cross entropy loss. A backbone network and two task-specific subnetworks make

up the single, coherent network known as RetinaNet [137]. The backbone is an off-the-shelf convolutional network that is in charge of calculating a convolutional feature map across an entire input image. It uses a backbone network like ResNet (ResNet-50/ResNet-101) for feature extraction. On top of this, a Feature Pyramid Network (FPN) is used to detect objects at different scales, enabling RetinaNet to handle multi-scale detection. Two sub-networks: a classification subnet for predicting the probability of objects and a bounding box regression subnet for locating objects. By addressing the issue of class imbalance, it achieves a high detection accuracy, especially for small and hard-to-detect objects [41]. RetinaNet uses FPN to handle objects of different sizes, making it more robust to scale variations in objects. The disadvantage of the network is, that RetinaNet's use of the FPN and focal loss can lead to more complex computations than other one-stage detectors like YOLO, resulting in slower inference and due to the feature pyramid and loss functions. RetinaNet can be memory-intensive, which can limit its use in resource-constrained environments. X. Cheng and J. Yu [42] propose RetinaNet for steel surface defect detection that uses adaptive spatial feature fusion and difference channel attention.

4.2.2.7. YOLOv3. You Only Look Once, version 3 (YOLOv3) is a widely adopted one-stage object detection model, and recognized for its real-time detection capabilities [137]. It builds on the architecture of the original YOLO and YOLOv2 models by improving both accuracy and speed. These advancements make YOLOv3 a popular choice for applications that demand fast and precise object detection. YOLOv3 uses a modified version of Darknet-53 as its backbone, which is a 53-layer convolutional neural network [88]. Darknet-53 is deeper and more powerful than previous YOLO backbones, allowing it to extract richer features from the input image. For class prediction, independent logistic classifiers are used, and binary cross-entropy loss is used instead of Softmax. For prediction, three different scales are used, and features are extracted from these scales [43]. YOLOv3 uses logistic regression to predict the presence of objects in each box for multiple classes. YOLOv3 uses the Leaky ReLU activation function for all convolutional layers except for the last one, where a linear activation is used to make the final predictions. It can process up to 45 frames per second (FPS) on a standard GPU, making it ideal for applications requiring fast inference. The use of feature pyramids allows YOLOv3 to detect objects of varying sizes with good accuracy. This makes it more robust in scenarios where objects appear at different scales. YOLOv3 improves on the accuracy of previous YOLO versions and is competitive with two-stage detectors in terms of performance. Jiang Xinbei et al. [44] propose a real-time face mask detection method based on YOLOv3.

4.2.2.8. CenterNet. CenterNet is an advanced object detection algorithm designed to simplify the object detection process by predicting the centre point of objects and their sizes instead of using anchor boxes or region proposals [137]. This approach allows it to be a highly efficient one-stage detector, often achieving both high accuracy and speed. One-stage object detector CenterNet recognizes each object as a triplet of key points instead of a pair [45]. It makes use of two specially designed modules called cascade corner pooling and centre pooling, which serve to improve the data gathered by the top-left and bottom-right corners and to provide more identifiable data in the middle areas, respectively. CenterNet typically uses popular backbones like Hourglass Network or ResNet for feature extraction. The hourglass architecture is effective at capturing multi-scale information, improving the detection of objects of different sizes. CenterNet generates a heatmap where each pixel represents the likelihood of it being the centre of an object. Higher values in the heatmap correspond to higher confidence in that region being an object centre. Along with the centre key point, CenterNet predicts the width and height of the bounding box. The offset to refine the centre point location more accurately. The network is trained to minimize the discrepancy between the actual and predicted

box dimensions, as well as the disparity between the ground truth and predicted heatmaps. CenterNet offers a clean and efficient approach to object detection by focusing on the core concept of keypoint prediction while avoiding some of the complexities inherent in anchor-based methods [46].

4.2.2.9. EfficientDet. EfficientDet is a family of object detection models built to optimize both speed and accuracy, leveraging the ideas from EfficientNet and the BiFPN (Bidirectional Feature Pyramid Network) [137]. EfficientDet uses EfficientNet as its backbone for feature extraction. EfficientNet is a highly optimized network that balances width, depth, and resolution scaling, providing better accuracy with fewer parameters compared to traditional networks like ResNet. EfficientDet introduces BiFPN, an enhanced version of the traditional FPN used in object detection tasks. BiFPN allows for a bidirectional flow of information across different scales of features, meaning that both top-down and bottom-up connections are optimized. Unlike standard FPN, BiFPN uses learnable weights to adjust the importance of features at each scale, which helps improve detection performance by prioritizing more relevant features [47]. EfficientDet also employs compound scaling, a technique that simultaneously scales the model's depth, width, and input resolution in a balanced manner. This allows the model to scale efficiently across different resource budgets, from lightweight models to high-performing large models. There are multiple variants of EfficientDet (D0 to D7) that scale up in terms of size and complexity, providing options depending on the trade-off between speed and accuracy: EfficientDet-D0 is the smallest and fastest version. EfficientDet-D7 is the largest and most accurate but also the slowest. Jia Jiaqi et al. [48] propose improved EfficientDet for underwater object detection.

4.2.2.10. YOLOv4, YOLOv5 and YOLOv7. YOLOv4 is a state-of-the-art object detector that outperforms all previous YOLO versions in terms of both accuracy and speed. It introduces a technique called "Bag of Freebies," which enhances training by leveraging data augmentation, Self-Adversarial Training, Cross Mini-Batch Normalization, CIoU-loss, DropBlock regularization, and a Cosine annealing scheduler, without impacting inference time [133]. The "Bag of Specials," which includes Mish activation, SPP-block, Multi-input Weighted Residual Connections, PAN Path-Aggregation Block, Cross-stage Partial Connections, and Spatial Attention Module block, is another technique that YOLOv4 uses to enhance inference performance. YOLOv4 employs a genetic method to choose hyperparameters and may be trained on a single GPU [51]. Shortly after YOLOv4's release, the company Ultralytics launched the YOLOv5 repository, offering significant improvements. YOLOv5, built with PyTorch, provides faster, easier, and more accurate deployment [79]. While YOLOv4 and YOLOv5 share similar frameworks, YOLOv5 has shown superior performance in certain scenarios. YOLOv5 offers five model variations: nano, small, medium, large, and extra-large [49].

YOLOv7 is a version of the YOLO series of object detection models that continues the tradition of providing high-speed and accurate detection [137]. It improves upon previous versions like YOLOv4 and YOLOv5 by introducing architectural changes and optimization techniques that make it more efficient and accurate [133]. YOLOv7 focuses on optimizing both speed and accuracy, making it one of the fastest and most accurate versions of YOLO available for object detection. YOLOv7 introduces E-ELAN (Extended Efficient Layer Aggregation Network) to efficiently aggregate and fuse features across different layers, improving the model's ability to learn more complex patterns [137]. YOLOv7 uses decoupled heads for classification and localization, meaning it separates the prediction of object classes and bounding boxes, leading to better performance in both tasks. YOLOv7 uses DLA (Dynamic Label Assignment) to improve the assignment of labels to predicted boxes during training, optimizing the learning process [50]. YOLOv7 incorporates techniques like a bag of freebies and a bag of specials to improve the model's generalization and training efficiency. YOLOv7 supports both

anchor-free and anchor-based detection, providing flexibility depending on the complexity of the dataset [133].

The comparison of YOLOv7 with other YOLO versions is as follows:

- YOLOv5: YOLOv7 surpasses YOLOv5 in terms of accuracy and speed, particularly for more complex object detection tasks. However, YOLOv5 remains easier to deploy and is widely adopted in many real-time applications [138].
- YOLOv4: YOLOv7 introduces more advanced training techniques and architectural innovations than YOLOv4, making it faster and more accurate. YOLOv4 is still efficient but is now largely overshadowed by YOLOv7 in performance [52].
- YOLOv3: YOLOv3 is significantly slower and less accurate than YOLOv7, though it is easier to implement on edge devices with fewer resources [138].

YOLOv7 is a state-of-the-art object detection model that combines speed, accuracy, and flexibility, making it suitable for a wide range of applications, from real-time video analysis to deployment on edge devices [137]. It represents a significant step forward in the YOLO family, particularly for tasks requiring both real-time detection and high precision.

The characteristics and the observation remarks of the one-stage models are given in Table 3.

4.2.3. Transformer based detectors

Transformers have significantly impacted the field of Natural Language Processing (NLP) since their inception [90], playing a key role in the development of state-of-the-art language models like Bidirectional Encoder Representations from Transformers (BERT) [91], Generative Pre-trained Transformer (GPT) [92], and Text-To-Text Transfer Transformer (T5) [93]. Utilizing the attention mechanism, transformers efficiently capture dependencies within sequences and handle longer contexts better than other sequential architectures. Building on their success in NLP, researchers began exploring transformers' potential in computer vision. The study in [94] introduced the Vision Transformer (ViT), the first transformer-based model tailored for image classification. Although Convolutional Neural Networks (CNNs) have traditionally dominated vision tasks, transformers face challenges, such as the absence of inductive bias and reliance on fixed post-training weights [95].

4.2.3.1. Detection transformer (DETR). DETR is a set-based object detector that applies a Transformer on top of a convolutional backbone [96]. It creates a 2D representation of the input image using a conventional CNN backbone, flattens it, adds positional encoding, and then feeds it into a transformer encoder. The transformer decoder handles the encoder's output after receiving a predetermined quantity of learnt positional embeddings, also known as object queries. A shared feed-forward network that predicts either a detection or a "no object" class is run through each decoder output embedding [134].

4.2.3.2. Vision Transformer (ViT). The Vision Transformer (ViT) has emerged as a strong alternative to convolutional neural networks (CNNs), which have long been the dominant architecture in computer vision for tasks like image recognition. ViT models surpass the current state-of-the-art CNNs in both computational efficiency and accuracy, offering nearly four times the performance [134]. While CNNs have traditionally led the field of computer vision, new Vision Transformer models have demonstrated exceptional capabilities, achieving results that are comparable to or even better than CNNs across a variety of computer vision tasks [53]. In ViTs, images are represented as sequences, and class labels for the image are predicted, which allows models to learn image structure independently.

Pengchuan Zhang et al. [54] propose a multi-scale vision longformer

for high-resolution image encoding. In ViTs, the image is first divided into patches, which are then flattened. These flattened patches are used to create lower-dimensional linear embeddings, which are combined with positional embeddings. The resulting sequence is then fed into a standard transformer encoder for training. ViT exhibits strong transfer learning capabilities, performing well on downstream tasks even with limited fine-tuning data. This makes it a strong candidate for tasks where labelled data is expensive or hard to obtain. While ViT has several advantages, it is worth noting that it requires large-scale pre-training to reach its full potential and may not perform as well as CNNs on smaller datasets without proper pre-training. However, with the increasing availability of large datasets and powerful computational resources, ViT represents a significant step forward in the field of computer vision [56].

4.2.3.3. Swim Transformer. A variation of the Vision Transformer [55], the Swim Transformer combines picture patches in deeper levels to provide hierarchical feature maps. Because self-attention is only calculated inside each local region, it provides linear computational cost concerning the input image size. Because of this, it can be used as a general-purpose backbone for tasks involving dense recognition and image classification. On the other hand, because self-attention is calculated globally, previous vision transformers have quadratic computational costs concerning the input image size and only provide feature maps at a single low resolution [134].

4.2.4. Lightweight networks

4.2.4.1. SqueezeNet. SqueezeNet is a convolutional neural network designed to reduce the number of parameters, primarily by using fire modules that "squeeze" parameters through 1×1 convolutions. SqueezeNet neural network had been created keeping AlexNet as the benchmark [57]. The model consists of convolution layers, fire modules and pooling layers. A squeeze convolution layer, consisting solely of 1×1 filters, is followed by an expanded layer that combines 1×1 and 3×3 convolution filters. The Fire module has three tuneable hyper-parameters: $s_{1 \times 1}$, $e_{1 \times 1}$, and $e_{3 \times 3}$. $s_{1 \times 1}$ represents the number of 1×1 filters in the squeeze layer, while $e_{1 \times 1}$ and $e_{3 \times 3}$ denote the number of 1×1 and 3×3 filters, respectively, in the expand layer [138]. The condition $s_{1 \times 1} < (e_{1 \times 1} + e_{3 \times 3})$ ensures that the squeeze layer limits the number of input channels to the 3×3 filters. As the name suggests, the squeeze layer serves to limit the number of inputs to the 3×3 filters, where 1×1 filters play a key role in dimensionality reduction Lee et al. [58] propose real-time vehicle make and model recognition with the residual SqueezeNet architecture.

4.2.4.2. MobileNet V, V2, V3. MobileNet is a convolutional neural network specifically designed for mobile and embedded vision applications. It utilizes depth-wise separable convolutions to significantly reduce the number of parameters and computations, making it more efficient than traditional CNNs [59]. A standard convolution operation combines both filtering and combining in a single step, while MobileNet splits this into depthwise convolution and pointwise convolution. Each input channel receives a single filter from the depthwise convolution, whereas the result of the depthwise convolution is combined over all channels by the pointwise convolution. This factorization significantly reduces computation [51]. Kalyani Kadam et al. [60] propose the detection and localization of multiple image splicing using MobileNetV1.

Except for using inverted residual blocks with bottlenecking features, MobileNetV2 is quite identical to the original MobileNet. Compared to the original MobileNet, it has a significantly smaller number of parameters. Any input size larger than 32×32 can be used with MobileNets; however, larger image sizes yield better results [138]. In contrast to conventional residual models, which employ extended representations in the input and output, the MobileNetV2 architecture is built on an

Table 3
Characteristics of one-stage models.

Sr. No.	Method	Year	Image Input Size	Region Proposal Method	Loss/ Cost Function	Remarks	Refs.
1.	DetectorNet	2017	Variable	FCN	Multi-task loss (classification + regression)	The model is efficient for object detection with fewer proposals and performs well in detecting smaller objects. However, it may struggle with larger objects and is less popular compared to other models like YOLO or SSD, which are more widely adopted in the field.	[111]
2.	OverFeat	2013	231×231	Sliding window approach with CNNs	Cross-entropy for classification, Smooth L1 loss for bounding box regression	As one of the earliest models to perform object detection with CNNs, it shows strong performance when fine-tuned for specific tasks. However, it is computationally expensive and has been outperformed by later models like YOLO and SSD, which offer better efficiency and accuracy.	[112]
3.	YOLO	2016	448×448	Single CNN that predicts bounding boxes and class probabilities.	Mean squared error for bounding boxes, cross-entropy for class predictions	The model is extremely fast, making it ideal for real-time applications, and supports end-to-end training. However, it has lower accuracy compared to region-based methods and struggles with detecting small objects, limiting its performance in certain scenarios.	[113]
4.	SSD	2016	300×300 or 512×512	Default box generation (multi-scale feature maps)	Softmax loss for classification, Smooth L1 loss for box regression	The model offers fast inference, making it suitable for real-time applications, and achieves high accuracy with multi-scale predictions. However, it struggles with detecting small objects and is less accurate compared to more complex models like Faster R-CNN.	[114]
5.	YOLOv2	2016	416×416 or 608×608	Single CNN, multiple anchor boxes	Multi-part loss (classification, bounding box regression)	The model is faster and more accurate than YOLO, with improvements for detecting small objects through the use of anchor boxes. However, it still struggles with very small objects compared to region-based models, which handle such cases more effectively.	[113]
6.	RetinaNet	2017	800×800	Feature Pyramid Networks (FPN) + anchor boxes	Focal loss (to address class imbalance)	The model achieves very high accuracy on challenging datasets and addresses class imbalance using focal loss. However, it is slower than single-shot detectors like YOLO and SSD and requires more computational resources, making it less suitable for real-time applications on resource-constrained devices.	[117]
7.	YOLOv3	2018	416×416 or 608×608	Single CNN with three detection scales	Multi-part loss (classification, bounding box regression)	The model offers better accuracy, especially for small objects, compared to previous YOLO versions, and provides faster inference with high mAP. However, it still falls short of the accuracy achieved by region-based methods like Faster R-CNN, particularly for more complex detection tasks.	[113]
8.	CenterNet	2019	512×512	Keypoint detection and center-ness score	Focal loss for classification and key point prediction	The model is simple and fast, excelling at detecting objects in crowded scenes. However, it struggles with objects that have large aspect ratios and has limited scalability for very large datasets, making it less suitable for certain complex detection tasks.	[118]
9.	EfficientDet	2020	512×512 , 640×640 , 768×768	EfficientNet backbone with BiFPN	Focal loss for classification, Smooth L1 loss for box regression	The model offers high efficiency and accuracy, making it optimized for mobile and edge devices. However, its complex architecture can make fine-tuning more challenging, requiring additional expertise for optimal performance.	[119]
10.	YOLOv4	2020	608×608	Single CNN with advanced techniques like PAN (Path Aggregation Network) and CSPDarknet53 backbone	Multi-part loss (classification, bounding box regression)	YOLOv4 offers significant improvements in accuracy over YOLOv3, with high speed and performance ideal for real-time detection. However, its heavier architecture requires more computational resources and is more complex compared to earlier YOLO versions, making it less accessible for resource-constrained environments.	[115]
11.	YOLOv5	2020	640×640	Single CNN with anchor boxes	Multi-part loss (classification, bounding box regression)	YOLOv5 is lightweight and fast, often offering better accuracy than YOLOv4 in many cases, and is easy to implement and highly customizable. However, it is not an official release from the original YOLO creators, despite its widespread adoption. Additionally, it still falls short of models like RetinaNet or Faster R-CNN in terms of accuracy on certain complex tasks.	[116]
12.	YOLOv7	2022	640×640	Single CNN with advanced optimizations like PANet and Mish activation function	Multi-part loss (classification, bounding box regression)	YOLOv7 delivers state-of-the-art performance with a strong balance of speed and accuracy, making it well-suited for real-time applications and general object detection tasks. However, it is	[120]

(continued on next page)

Table 3 (continued)

Sr. No.	Method	Year	Image Input Size	Region Proposal Method	Loss/ Cost Function	Remarks	Refs.
						computationally demanding and, being a newer model, may have fewer resources or community support compared to earlier YOLO versions.	

inverted residual structure, where the input and output of the residual block are thin bottleneck layers. In the intermediate expansion layer, MobileNetV2 filters features using lightweight depth-wise convolutions [61]. Wibowo Adi et al. [62] propose android skin cancer detection and classification based on the MobileNetV2 model.

MobileNetV3 is optimized for mobile phone CPUs using a combination of hardware-aware Network Architecture Search (NAS) and the NetAdapt algorithm [138]. It further enhances performance through architectural advancements, including complementary search methods, efficient nonlinearities tailored for mobile use, and innovative network designs. MobileNet models are known for achieving high accuracy while minimizing size and computation. For instance, MobileNetV3 [64] achieves similar performance to larger models like ResNet while being much more efficient. Bi Chunguang et al. [63] propose an identification method for corn leaf disease based on an improved MobileNetV3 model.

4.2.4.3. ShuffleNet. ShuffleNet is another efficient convolutional neural network architecture designed for mobile and embedded devices with limited computational resources, similar to MobileNet [138]. It was introduced by the researchers at Megvii in 2018. ShuffleNet primarily focuses on optimizing computation for devices with low power consumption and limited processing capabilities, such as smartphones. To reduce computation without compromising accuracy, it incorporates two innovative techniques: channel shuffle and pointwise group convolution. Like residual blocks in ResNet, ShuffleNet uses ShuffleNet units, which contain group convolutions, depthwise separable convolutions, and channel shuffling, among other components. These units stack to form the architecture of the network. Each ShuffleNet unit typically consists of three key operations, first Pointwise Group Convolution which reduces computational cost by using group convolutions, secondly, Channel Shuffle which shuffles the channels for better information flow across groups and third operation is Depthwise Convolution which is like MobileNet, used to reduce computation [140]. ShuffleNet has two variants i.e. ShuffleNetV1 and ShuffleNetV2. ShuffleNetV1 employs group convolutions and channel shuffling to build an efficient network. While it achieves high efficiency, it faces certain limitations related to complexity and speed. Tesfai H. et al. [66] propose lightweight ShuffleNet-based CNN for arrhythmia classification.

ShuffleNetV2 is an improved version focusing on practical efficiency rather than just theoretical efficiency [65]. ShuffleNetV2 is an improved version of ShuffleNetV1, as it has simplified architecture by removing unnecessary operations, making it faster. Secondly, It has a balanced computational cost means ensuring that each operation in the network takes a similar amount of time, avoiding bottlenecks. The third improvement focuses on reducing memory access costs. The model architecture is designed to minimize these costs, which significantly enhances performance on real-world hardware.

4.2.4.4. PeleeNet. PeleeNet is a convolutional neural network and object detection backbone that is a variation of DenseNet with optimizations to meet a memory and computational budget. PeleeNet incorporates techniques inspired by SqueezeNet, which focuses on reducing the number of parameters while maintaining accuracy [140]. It uses squeeze layers to limit the number of input channels, followed by expand layers to increase the number of output channels selectively. This compression-expansion strategy is effective in reducing model size without sacrificing much performance. PeleeNet is specifically optimized for low memory usage [138]. Its architecture is designed to

enable deployment on devices with limited RAM, making it highly suitable for real-time applications where both latency and memory footprint are critical constraints. Pinto A. et al. [68] uses PeleeNet for scene text detection. When comparing the PeleeNet with the MobileNet, it is observed that both PeleeNet and MobileNet use depthwise separable convolutions to reduce computation, but PeleeNet often focuses more on reducing the number of parameters and can achieve an even smaller model footprint in some cases. While compared with SqueezeNet, it is observed that PeleeNet borrows concepts from SqueezeNet, particularly the idea of compressing and expanding channels in a way that reduces the model size without greatly affecting performance. While comparing with ShuffleNet, it is concluded that ShuffleNet focuses on group convolutions and channel shuffling, while PeleeNet emphasizes a more straightforward use of depthwise separable convolutions and channel squeezing [67].

4.2.4.5. ESPNet. Efficient Spatial Pyramid Network (ESPNet) is a highly efficient convolutional neural network architecture designed for semantic segmentation tasks. ESPNet is 22 times faster on a standard GPU and 180 times smaller than the state-of-the-art semantic segmentation network PSPNet [138]. It employs ESP modules for both convolutional kernel learning and down-sampling operations, except in the first layer, which uses a standard stride convolution [140]. Each layer, except the final one, is followed by batch normalization and a PReLU activation function. The final layer feeds into a Softmax for pixel-wise classification. ESPNet has three variants designed to enhance segmentation performance: ESPNet-A, ESPNet-B, and ESPNet-C. ESPNet-A serves as the standard version, processing an RGB image using ESP modules to learn multi-level spatial representations and produce a segmentation mask [70]. ESPNet-B improves upon this by enhancing information flow through feature map sharing between the previous stride ESP module and the preceding ESP module. ESPNet-C further refines ESPNet-B by incorporating the input image directly into the network to enhance information flow. All three variants generate outputs with spatial dimensions reduced to 1/8 of the input image. It is highly efficient in terms of both computation and memory and optimized for real-time semantic segmentation on devices with limited resources. Arora, S. et al. [69] propose advancing spoken language understanding through ESPNet.

4.2.4.6. MnasNet. MnasNet is a convolutional neural network designed for mobile devices, discovered using Mobile Neural Architecture Search. This approach explicitly includes model latency in its objective, enabling the identification of a network that balances accuracy and latency effectively. One of its key features is the use of inverted residual blocks, which are borrowed from MobileNetV2, helping it maintain high accuracy with low computational costs [72].

By factorizing a CNN model into distinct blocks and then searching for the operations and connections in each block independently, MnasNet employs a novel factorized hierarchical search space that permits various layer architectures in distinct blocks [138]. The mnasNet-A1 model consists of a variety of layer architectures throughout the network. MnasNet uses both 3×3 and 5×5 convolutions, which is different from previous mobile models that all only use 3×3 convolutions [71].

4.2.4.7. ProxylessNet-Mobile. ProxylessNet-Mobile is a convolutional neural network architecture developed using the ProxylessNAS neural

architecture search algorithm, specifically optimized for mobile devices [73]. Similar to other mobile-optimized networks such as MobileNetV2, ProxylessNet-Mobile employs inverted residual blocks. These blocks help reduce computational costs while preserving the network's ability to capture complex patterns [138]. The ProxylessNet-Mobile variant specifically focuses on finding an architecture that is efficient for mobile devices by searching directly for architectures that minimize both latency and memory footprint without compromising performance [74]. ProxylessNet-Mobile provides a solution that balances accuracy, latency, and memory usage, making it a strong choice for mobile AI tasks. It achieves this by conducting a hardware-aware neural architecture search directly on the target hardware, leading to a highly optimized network for deployment on mobile devices [140].

4.2.4.8. FBNet. FBNet is a type of convolutional neural architecture discovered through differentiable neural architecture search (DNAS) [75]. FBNet uses a differentiable search process. This allows the network architecture to be optimized end-to-end using gradient-based methods, making the search process more efficient. The key innovation is that the architecture is represented as a super network where each layer can be one of several candidate operations. The optimal architecture is chosen by applying gradients to the operations in a continuous manner, enabling fast and efficient search [138]. FBNet is notable for its use of DNAS, which allows it to directly optimize for both accuracy and latency on specific hardware, such as smartphones, with great efficiency. It utilizes a basic type of image model block inspired by MobileNetV2 that utilizes depthwise convolutions and an inverted residual structure [140]. FBNet comes in several variants i.e. FBNet-A, FBNet-B and FBNet-C, each optimized for different performance levels. These variants offer different trade-offs between accuracy and latency, making it easy to select the right model for specific use cases [76]. For instance, FBNet-A is the smallest and fastest model, optimized for scenarios where latency is critical, while FBNet-C offers better accuracy at a slight cost in latency.

4.2.4.9. GhostNet. GhostNet is a convolutional neural network designed with Ghost modules to generate more features while using fewer parameters [140]. It consists primarily of Ghost bottlenecks, which serve as the building blocks of the network. The architecture begins with a standard convolutional layer with 16 filters, followed by a series of Ghost bottlenecks with progressively increasing channels, grouped into stages based on the input feature map sizes. Most Ghost bottlenecks use a stride of 1, except for the final one in each stage, which uses a stride of 2 [78]. The network concludes with global average pooling and a convolutional layer, producing a 1280-dimensional feature vector for classification. Some Ghost bottlenecks incorporate a squeeze-and-excite (SE) module in their residual layers [138]. Unlike MobileNetV3, GhostNet avoids the hard-swish activation function due to its high latency. Paoletti, M.E. et al. [77] used GhostNet for hyperspectral image classification.

4.2.4.10. MobileDet. MobileDet is a lightweight object detection model designed specifically for mobile and edge devices. It combines the advantages of efficient neural network architectures and advanced object detection techniques to provide a good balance between speed, accuracy, and computational efficiency [138]. MobileDet is based on the idea of utilizing resource-efficient building blocks, such as depthwise separable convolutions and attention mechanisms, to achieve high performance with minimal computational overhead [80]. MobileDet is built on top of a lightweight backbone such as MobileNetV3, EfficientNet, or other efficient architectures designed for mobile devices. The detection head is optimized to process the extracted feature maps from the backbone and detect objects in various scales and locations [139,140]. To improve object detection, MobileDet integrates attention mechanisms that help the model focus on relevant regions of the image, while

still maintaining efficiency. The advantages of MobileDet are, that it is optimized to run on devices with limited computational resources, such as smartphones, drones, and IoT devices, while maintaining real-time object detection capabilities and the model is designed for low-latency inference, which is crucial for real-time applications [81].

4.2.4.11. MobileViT. Mobile Vision Transformer (MobileViT) is a hybrid model that combines the strengths of CNNs and transformers to achieve efficient object detection, particularly on mobile or resource-constrained devices [136]. It blends the lightweight and efficient nature of CNNs with the strong attention mechanisms of transformers, leading to improved performance on tasks like object detection, segmentation, and classification [82]. MobileViT utilizes both convolutional layers and transformer layers in its architecture i.e. Convolutional Layers for local feature extraction and Transformer Blocks for capturing global dependencies and enhancing contextual understanding, which is particularly useful for object detection [137]. The core of MobileViT is designed to replace some of the standard convolutional blocks in CNNs with transformer blocks, where feature maps from the convolutional layers are fed into a transformer for attention-based refinement [138]. The model's key features include a lightweight architecture, efficient feature representation, and scalability. It can be easily adapted to different model sizes, ranging from compact versions for mobile devices to larger versions for server-based applications [139]. MobileViT offers a compelling balance between accuracy and efficiency, making it ideal for deployment in scenarios where resources are limited but real-time object detection is critical. Liu, J. et al. [83] propose an anti-noise bearing fault diagnosis method using adaptive deconvolution and MobileViT.

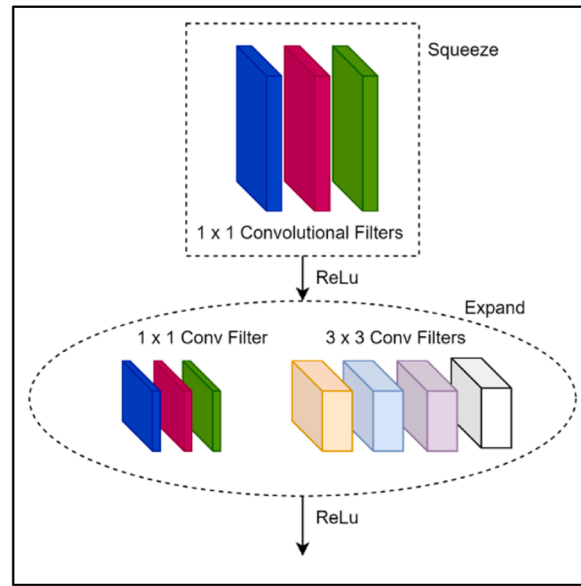
4.2.4.12. MoviNet. Mobile Video Networks (MoviNet) is a state-of-the-art neural network architecture designed by Google Research for real-time video processing, specifically optimized for mobile devices and streaming applications [140]. MoviNet is designed to process video frames efficiently and can be employed in object detection by leveraging temporal information from consecutive frames to improve detection accuracy [139]. The MoviNet models are designed to work efficiently on mobile and edge devices, providing high performance with lower computational costs [84]. It supports "streaming inference", meaning it can process video frames sequentially rather than having to process entire video sequences at once. MoviNet combines spatial information with temporal information to enhance understanding in video contexts. These models are optimized for real-time applications, making them suitable for scenarios where rapid object detection is needed [85]. The streaming architecture allows for continuous inference on live video feeds without the need to wait for entire video sequences, which reduces delay in real-time systems. Compared to conventional models, MoviNet is highly efficient, requiring fewer FLOPs (floating-point operations per second) to achieve competitive accuracy [138].

The architectures of the Lightweight models are presented in Fig. 6 and the discussion on characteristics of the Lightweight models which includes input image size, loss function etc. are highlighted in Table 4.

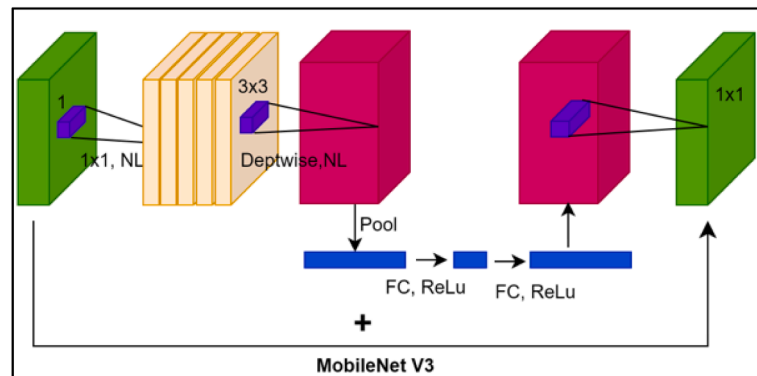
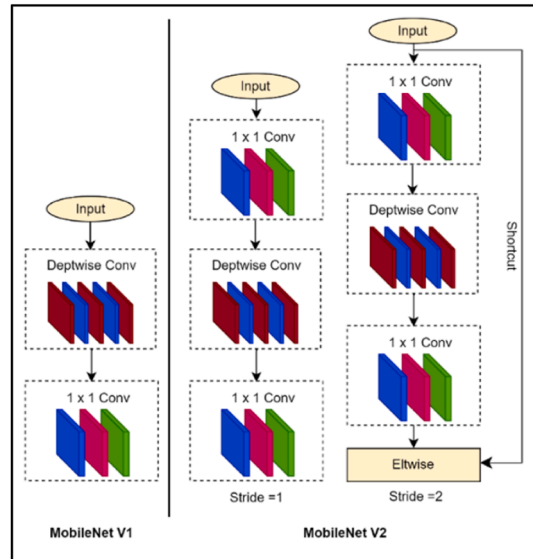
5. Evaluation metrics

The effectiveness of object detectors can be evaluated using various metrics, including Accuracy, Precision, Intersection over Union (IoU), Recall, Precision-Recall (PR) curve, and Average Precision (AP). Among these, Average Precision is the most commonly used metric, as it is derived from both recall and precision [87].

The main objective of object detectors is to locate objects in an image or video by drawing bounding boxes around them while associating each box with a specific class and a high confidence score [97]. Object detection involves three core components: the object class, the bounding box enclosing the object, and the confidence score [100]. The metrics used to evaluate the performance of object detection algorithms are



(a)



(b)

Fig. 6. Architecture and Block Diagrams of Lightweight Models (a)SqueezeNet (b)MobileNetV, V2, V3 (c) ShuffleNet (d) PeleeNet (e)ESPNet (f) MnasNet (g) FBNet (h) GhostNet (i) MobileViT.

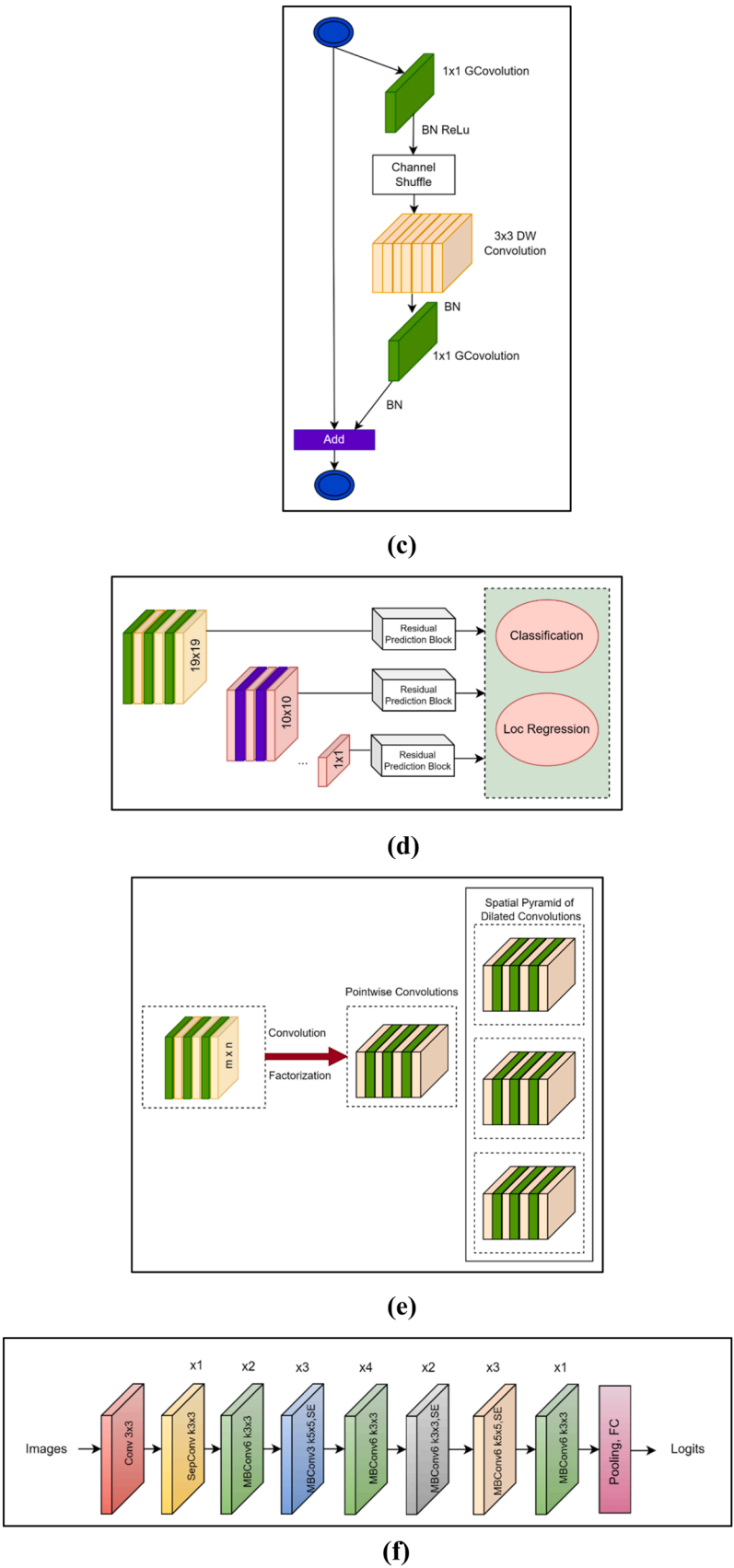


Fig. 6. (continued).

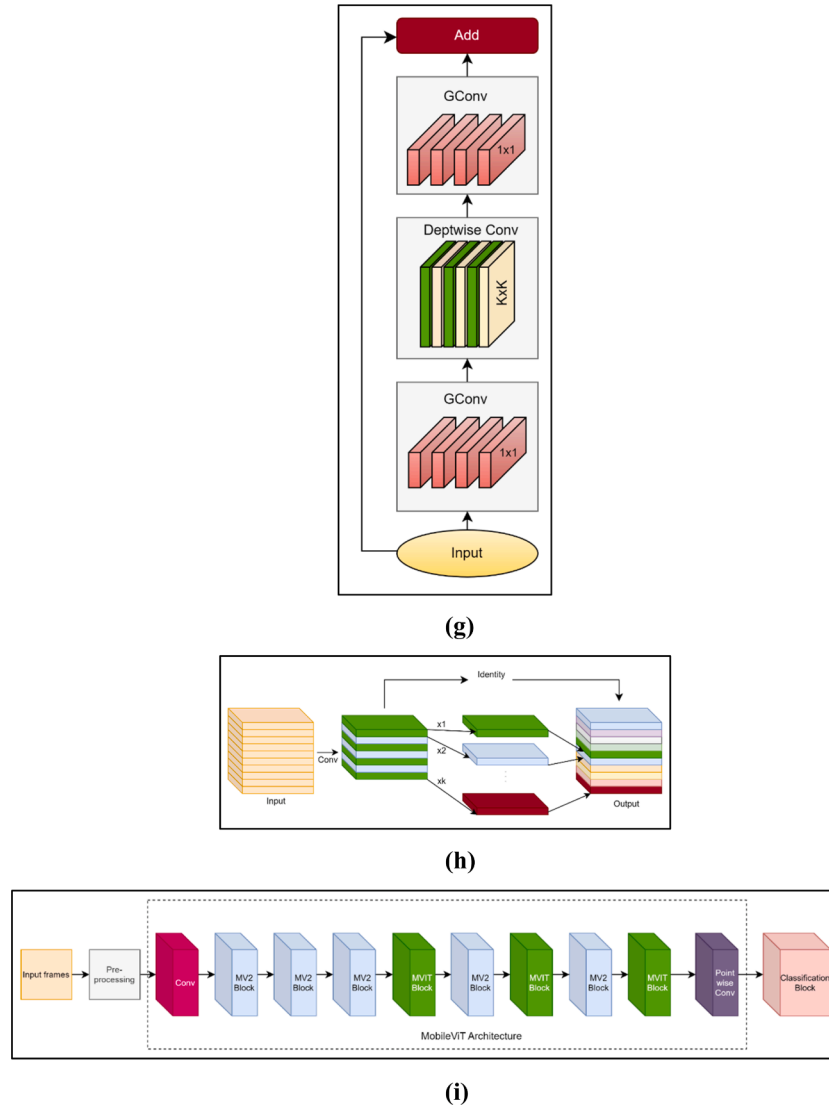


Fig. 6. (continued).

described below:

5.1. IoU (Intersection over Union)

IoU is a metric that quantifies the overlap between the predicted bounding box and the ground truth bounding box relative to the area of their union. It is based on the Jaccard index, which measures the similarity between the two sets [98]. The IoU value ranges from 0 to 1, with higher values indicating more accurate detections. An IoU of 1 signifies a perfect overlap between the predicted and ground truth bounding boxes, while an IoU of 0 indicates no overlap. If the IoU value exceeds a pre-defined threshold, typically set at 0.5, the object is considered correctly detected. IoU is calculated as follows:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

For each object detection task, precision and recall are evaluated based on the IoU value for a given threshold (T). When $IoU \geq T$, predictions are deemed correct; otherwise, if $IoU < T$, predictions are classified as incorrect [139]. To compute precision and recall, each bounding box must be categorized as follows:

True Positive (TP): The model correctly predicts the object (positive) and it is present.

True Negative (TN): The model correctly predicts the absence of the object (negative) and it is indeed absent.

False Positive (FP): The model incorrectly predicts the presence of the object (positive) when it is not present.

False Negative (FN): The model fails to detect the object (negative) when it is actually present.

5.2. Accuracy

It shows how well the model performs overall across all classes. It is computed by dividing the total number of samples that were successfully classified by the total number of samples [100,139]. The formula is as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

5.3. Precision

Precision measures the proportion of correct positive predictions. It is calculated as the ratio of correctly identified positive samples to the total number of predicted positive samples [101,139].

Table 4
Characteristics of lightweight models.

Sr. No.	Model	Year	Input Image Size	Loss Function	Remarks	Refs.
1.	SqueezeNet	2016	224 × 224	Cross-Entropy Loss	The model is extremely lightweight, with fewer parameters, making it ideal for deployment on devices with limited computational resources. Despite its small size, it offers high accuracy relative to its scale. However, its performance is lower compared to larger models like ResNet, and it offers less flexibility for transfer learning tasks, limiting its adaptability to more complex problems.	[121]
2.	MobileNet V1	2017	224 × 224	Cross-Entropy Loss or SSD Loss (For detection tasks)	Depthwise separable convolutions help reduce computation, making the model highly efficient and well-suited for mobile devices. While it delivers excellent performance on such devices, its accuracy is slightly lower compared to larger models like Inception or ResNet, which may offer better results on more complex tasks.	[122]
3.	MobileNet V2	2018	224 × 224	SSD Loss, Focal Loss (for detection tasks)	The model introduces inverted residuals and linear bottlenecks, enhancing efficiency while maintaining a small size. It offers improved accuracy compared to its V1 version with minimal increase in size. However, it still lags behind larger models in terms of overall accuracy, particularly for more complex tasks.	[123]
4.	MobileNet V3	2019	224 × 224	SSD Loss, Focal Loss	Optimized using neural architecture search (NAS), the model achieves improved accuracy and efficiency compared to its V1 and V2 versions. However, its complex architecture demands more expertise for fine-tuning, making it less accessible for users with limited experience.	[124]
5.	ShuffleNet	2018	224 × 224	Cross-Entropy Loss	The model uses channel shuffling to reduce computation, making it lightweight and efficient for mobile devices. However, its limited flexibility in handling large-scale tasks may hinder performance on more complex problems.	[125]
6.	PeleeNet	2018	224 × 224	SSD Loss	The model combines dense connections and inception modules, offering a lightweight design with a good balance of speed and accuracy. However, it is less popular and lacks extensive pre-trained weights, limiting its ease of use and adoption in some tasks.	[126]
7.	ESPNet	2018	Variable	Pixel-wise Cross-Entropy Loss (for segmentation tasks)	The model is extremely lightweight, making it well-suited for segmentation tasks, and features an efficient spatial pyramid for multi-scale processing. However, its accuracy is lower compared to more advanced models like DeepLab, which may offer better performance on complex segmentation challenges.	[127]
8.	ProxylessNet-Mobile	2019	Variable	Cross-Entropy Loss	Optimized using neural architecture search (NAS) without proxy models, the model strikes a balance between accuracy and latency. However, the NAS process requires significant computational resources for training, making it resource-intensive and less accessible for some users.	[74]
9.	FBNet	2019	224 × 224	Cross-Entropy Loss	The NAS-designed lightweight model offers high efficiency and excellent latency on edge devices, making it ideal for resource-constrained environments. However, the architectures generated through NAS can be difficult to interpret, posing challenges in understanding and modifying the model for specific use cases.	[128]
10.	GhostNet	2020	224 × 224	Cross-Entropy Loss	By introducing Ghost modules, the model reduces computation, resulting in a lightweight and efficient design for mobile applications. However, its accuracy is slightly lower than that of larger architectures, which tend to perform better on more complex tasks.	[129]
11.	MobileDet	2020	Variable	SSD Loss, Focal Loss	Optimized for detection on resource-constrained devices, the model offers an excellent balance of speed and accuracy. However, its performance on large-scale datasets is less competitive compared to more complex models, which can handle larger datasets with higher accuracy.	[130]
12.	MobileViT	2021	Variable	Cross-Entropy Loss, Focal Loss	The model combines vision transformers with mobile efficiency, offering high accuracy across a range of tasks. However, it is computationally heavier than purely convolutional models, which may limit its performance on devices with fewer resources.	[82]
13.	Movinet	2021	Variable	Cross-Entropy Loss	Optimized for video-based tasks with efficient temporal modelling, the model delivers good accuracy for real-time applications. However, its specialized design limits its application outside of video tasks, making it less versatile for other types of problems.	[84]

$$\text{Precision} = \frac{TP}{TP + FP}$$

5.4. Recall

Recall quantifies how many actual positive instances are correctly identified. It is calculated as the proportion of accurately identified positive samples to all positive samples. Recall is also referred to as sensitivity [101].

$$\text{Recall} = \frac{TP}{TP + FN}$$

5.5. Average Precision (AP)

Average Precision (AP) is one of the most commonly used metrics to evaluate detection accuracy. It is computed separately for each object

category. AP evaluates the precision-recall curve by computing the area under it [101].

$$AP = \int_0^1 P(R) dR$$

$$P(R) = \text{Precision}$$

5.6. Mean Average Precision (mAP)

Mean Average Precision (mAP) is calculated by averaging the AP values across all object categories, providing a comprehensive assessment of an object detector's performance.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

N = Number of classes

AP_i = Average Precision for class i

5.7. F1-Score

The F1-Score represents the balance between precision and recall [139]. It is the harmonic mean of these two metrics and is especially useful in cases of imbalanced class distribution. High precision and recall lead to a high F1-Score, while a low value indicates a significant imbalance between the two [99].

$$F1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5.8. Precision-recall curve (PR curve)

The precision-recall curve illustrates the trade-off between recall and precision at different thresholds. A larger area under the curve indicates better performance with high recall and precision. PR curves are particularly informative when evaluating binary classifiers on imbalanced datasets. As recall increases, if precision remains high, the detector performs well. Conversely, if recall decreases while precision remains high, maintaining precision at a certain level is essential to balance recall. PR curves are widely used in research for positive prediction analysis [139].

5.9. AUC-ROC curve

AUC, or Area Under the ROC Curve, evaluates the performance of classification models across various thresholds. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR). A larger area under the ROC curve signifies better performance, with an ideal curve approaching the upper-left corner (0, 1). AUC values range from 0.5 to 1, with higher values indicating more accurate detection performance [139].

6. Comparative analysis of object detection algorithms based on strengths and weaknesses

This section presents a concise yet insightful analysis of prominent object detection algorithms, evaluating their strengths and limitations. Traditional object detection methods laid the foundation for detecting objects in images. However, their limited adaptability to complex scenarios and variations in object appearances led to the rise of deep learning methods that revolutionized the field. Deep learning models have achieved state-of-the-art performance across a variety of benchmarks. These models benefit from their ability to learn hierarchical features, significantly improving accuracy and robustness. However, they often require high computational resources and large datasets, which can limit their deployment on resource-constrained devices. One-stage models, process images in a single pass, making them highly efficient for real-time applications. They are characterized by their fast inference speed and lower computational cost, making them ideal for tasks like autonomous driving and video surveillance. However, they often struggle with detecting small or overlapping objects and tend to produce more false positives due to dense predictions. In contrast, two-stage models, follow a more refined approach by first generating region proposals before classifying objects, resulting in superior accuracy and robustness in complex scenes. These models excel in handling small objects and reducing false detections but suffer from slower inference times and higher computational demands, making them less suitable for real-time applications. While one-stage models are preferred for speed-critical tasks, two-stage models are better suited for scenarios where precision is paramount, such as medical imaging and high-precision security systems. Transformer-based models have introduced a paradigm shift by eliminating the need for region proposal networks and

leveraging attention mechanisms. However, transformers generally require significant computational power and memory, which limits their deployment in real-time and edge applications. The emergence of lightweight models designed specifically for edge devices and mobile applications addresses the challenge of balancing computational efficiency with accuracy. These models optimize architectures for reduced complexity while maintaining competitive performance, making them more suitable for real-time and embedded systems. Lightweight models offer a compromise between accuracy and efficiency but may underperform on highly complex datasets compared to larger models.

The best object detection algorithms, considering their accuracy, speed, and suitability for different applications are explained in Table 5. As given in Table 5 YOLOv5/YOLOv7, Faster R-CNN, EfficientDet, SSD, RetinaNet, and DETR are the best algorithms for different applications as per our research survey.

Each algorithm has its strengths, making some ideal for real-time applications and others for high-precision tasks. The choice depends on the computational resources available and the specific requirements of the respective application.

7. Conclusion, future directions, and key challenges

This review paper provides a thorough analysis of state-of-the-art object detection models i.e. Traditional models and Deep Learning models which include two-stage detectors, one-stage detector, transformer-based detectors, and lightweight networks and explain evaluation metrics used to evaluate the performance of models. The architectures of the models are thoroughly explained, providing a comprehensive analysis of their structural intricacies. Additionally, a comparative summary is presented in tabular form, encapsulating the defining characteristics of various model categories, and facilitating a clear and insightful evaluation of their differences and similarities. The performance metrics used to evaluate object detection models are systematically discussed in this paper. Towards the end, the comparative analysis of object detection algorithms based on strengths and weaknesses, best object detection models for various applications, future directions and key challenges are highlighted based on the study. The survey reveals that, despite significant advancements and remarkable performance in object detection methods, there remains considerable scope for further improvement.

7.1. Future directions and key challenges

Despite remarkable progress in object detection, several critical challenges remain and require urgent attention. One of the primary difficulties is achieving the optimal balance between accuracy, speed, and computational efficiency. High-performance models often demand significant processing power, making them unsuitable for real-time applications and resource-constrained environments. Another major

Table 5
Best object detection algorithms.

Sr. No.	Algorithm	Strengths	Best For
1.	YOLOv5/YOLOv7	Real-time detection, mobile efficiency	Mobile apps, real-time video, drones, autonomous driving
2.	Faster R-CNN	High accuracy, small object detection	Healthcare, precision-demanding tasks, autonomous vehicles
3.	EfficientDet	Scalable, efficient	Applications requiring a trade-off between accuracy and speed
4.	SSD	Fast, simple	Mobile devices, real-time object detection
5.	RetinaNet	Focal Loss for small objects	Small object detection in satellite images, drones
6.	DETR	Transformer-based, global context	Complex scenes, video analytics

challenge is ensuring robustness and generalization, as many models struggle to maintain consistent performance under varying conditions such as changes in lighting, occlusions, and complex backgrounds. Enhancing model adaptability without extensive retraining remains a crucial research focus. Furthermore, small object detection continues to be problematic due to limitations in feature extraction and resolution. Advancing multi-scale feature representation and incorporating more sophisticated architectures could help address this issue. Lastly, the scarcity of annotated data in many domains poses a significant hurdle. Leveraging self-supervised learning, few-shot learning, and data-efficient training techniques will be essential to overcoming this limitation. Addressing these challenges is key to advancing the next generation of object detection models, making them more accurate, efficient, and adaptable for real-world applications.

CRediT authorship contribution statement

Vrushali Pagire: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Murthy Chavali:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Ashish Kale:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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